Persistent Homology of UK Socioeconomic Life-Course Trajectories

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Abstract

Scalar persistence summaries and diagram-level Wasserstein distances give opposite verdicts on whether first-order Markov dynamics account for the topology of UK life-course trajectories. We apply persistent homology (PH) to 27,280 employment-income trajectories drawn from Understanding Society (2009–2023), with cross-era validation on 8,509 trajectories from the British Household Panel Study (1991–2008). Each trajectory is a sequence of nine states combining employment status (employed, unemployed, inactive) with income band (low, mid, high), embedded into 20-dimensional PCA space via n-gram frequency vectors and analysed using Vietoris–Rips persistent homology with maxmin landmarks. A **Markov memory ladder**—five null models of escalating structural complexity—characterises the generating process underlying the observed topology.

Seven distinct mobility regimes emerge via Gaussian mixture models optimised by BIC. Total-persistence tests suggest that temporal ordering creates topological structure (order-shuffle H_0 $p < 0.005$) while first-order Markov dynamics fully account for the topology (Markov-1 H_0 $p = 1.000$). Upgrading the test statistic to Wasserstein diagram-level distances—comparing the full geometric configuration of persistence features rather than collapsing them to a single scalar—reverses the Markov finding. Under Wasserstein testing (100 permutations, 2,000

landmarks), the Markov-1 null is rejected for H_0 ($p = 0.002$): observed persistence diagrams differ significantly in shape from Markov-generated diagrams, even though Markov surrogates produce more total persistence. Order-shuffle H_0 is borderline under Wasserstein ($p = 0.058$), label shuffle serves as a negative control ($p = 0.452$), and H_1 (loops) remains non-significant under all null types. The discrepancy between scalar and diagram-level tests is itself a methodological finding: total persistence masks structural differences that Wasserstein distances detect.

These findings replicate on BHPS-era trajectories (1991–2008): order-shuffle H_0 $p = 0.000$, Markov-1 H_0 $p = 0.000$—with the BHPS data additionally rejecting Markov-1 H_1 ($p = 0.000$). Stratified Wasserstein distances reveal significant topological differences by gender, parental NS-SEC, and birth cohort; 30/50 tests survive Benjamini–Hochberg FDR correction at $q < 0.05$. Overlapping 10-year career-phase windows confirm extreme regime stickiness: among 7,453 individuals starting in disadvantaged regimes, only 416 (5.6%) escape to advantaged regimes (11.0% under non-overlapping windows). Comparison with optimal matching (OM) baselines (ARI = 0.26 at matched $k = 7$) confirms that TDA's primary added value lies in formal null testing and multi-scale geometric characterisation.

Keywords: persistent homology, topological data analysis, social mobility, life-course trajectories, Markov models, regime switching, Wasserstein distance

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1 Introduction

1.1 Motivation: UK Social Mobility and Life-Course Dynamics

Intergenerational social mobility in the United Kingdom has stagnated or declined since the 1970s: recent cohorts show lower rates of upward movement relative to their parents' generation (Blanden et al., 2004, 2013; Goldthorpe and Mills, 2008). Behind this stagnation lie persistent unemployment, income churning, and structural inactivity — dynamics that create multiple equilibria in employment-income states. Cross-sectional snapshots — comparing parents' and children's occupational class or income at a single point — often miss these dynamics. What matters is not merely where individuals start and end, but the shape of the path between.

The British Household Panel Study (BHPS, 1991–2008) and its successor Understanding Society (USoc, 2009–2023) together provide two of the longest continuous household panels in the social sciences. Our primary analysis draws on the USoc-era panel: from over 118,000 respondents, we reconstruct 27,280 individual trajectories with at least 10 consecutive years of data within the USoc observation window (start years 2009–2013). Each trajectory is a time-ordered sequence through a nine-state space combining employment status — employed (E), unemployed (U), or inactive (I) — with income band — low (L), mid (M), or high (H) — derived from equivalised household income relative to 60% of the contemporary median. As cross-era validation, we separately reconstruct 8,509 BHPS-era trajectories (1991–2008, mean length 14.5 years) by fixing the BHPS `{wave}pid` → USoc `pidp` linkage via the cross-wave identifier file, and equivalising BHPS raw household income (`fihhmn`) using the modified OECD scale (`eq_moecd`). We also identify 8,459 spanning trajectories observed in both surveys, enabling 20+ year longitudinal tracking (§4.7).

The resulting trajectories reveal a labour market in which employment dominates (58% of person-years) but inactivity is pervasive (32%), and low income affects over a third (36%) of observed person-years. These are not transient states: the modal trajectory length in our sample is 13 years, sufficient to observe structural persistence,

cycling, and — for some — escape from disadvantaged states.

UK labour market trajectories are also shaped by regional economic structures and local labour market conditions (HM Government, 2022). The present analysis operates at the national level; regional stratification (NUTS2 or Travel-to-Work Area) is a natural extension of the Markov memory ladder (§5.4).

The policy question is whether disadvantaged trajectories are absorbing or transient. If low-income churning creates persistent cycles from which exit is rare, marginal interventions targeting single transitions will not be enough. If trajectories are well described by low-order Markov dynamics, targeting the right transitions could suffice. Distinguishing between these scenarios requires methods that go beyond measuring distances between sequences, and — as we show — test statistics that compare persistence diagrams as geometric objects rather than collapsing them to scalars.

1.2 Research Gap

Sequence analysis methods — optimal matching (OM) and its many variants — have dominated life-course research since Abbott's (1995) influential importation of biological sequence techniques into sociology. OM and related methods treat trajectories as strings and compute pairwise distances under substitution and indel (insertion-deletion) cost matrices, enabling cluster analysis to identify "types" of life course. This approach has been productive, generating typologies of employment careers, family formation, and housing trajectories across many national contexts (Abbott and Tsay, 2000; Lesnard, 2010; Gauthier et al., 2010).

However, sequence analysis has limitations for the questions we pose. OM distances depend on the substitution cost matrix, which is typically set a priori or estimated from transition rates — introducing subjectivity at a foundational level (Robette and Thibault, 2008). OM and its variants also treat trajectories as one-dimensional objects: they measure pairwise distances but have no concept of the global geometry or topology of the trajectory space. They cannot naturally detect persistent clusters (connected components), loops (cyclical traps), or higher-dimensional voids in the point

cloud of embedded trajectories — though sequence-based clusters may capture similar substantive phenomena by other means. Nor is there a standard framework for formal significance testing — for asking whether the observed structure exceeds what would be expected under explicit generative null models.

Markov chain models, the other workhorse of life-course analysis, capture local transition probabilities but are not designed to detect global shape. A Markov model can tell us the probability of moving from unemployment to employment, but it cannot detect whether the cloud of all trajectories in embedding space has a topological signature of cyclical trapping, or whether the number and separation of trajectory clusters exceeds what memoryless transitions would produce.

Topological data analysis (TDA), via persistent homology (PH), addresses these limitations. PH identifies topological features — connected components (H_0), loops (H_1), and higher-dimensional voids — that persist across a range of spatial scales, providing a multi-scale signature of the shape of a point cloud that is robust to noise and dimensionality (Carlsson, 2009; Edelsbrunner and Harer, 2010). TDA has been applied to financial time series (Gidea and Katz, 2018), brain networks (Saggar et al., 2018), gene expression (Rizvi et al., 2017), and materials science (Hiraoka et al., 2016). PH hypothesis testing has been developed by Robinson and Turner (2017), and TDA–Markov hybrids have appeared in network analysis (Stolz et al., 2017). Recent work has brought TDA to adjacent social-science settings — notably network persistence in organisational sociology (Sizemore et al., 2018) and simplicial complexes for modelling social contagion (Iacopini et al., 2019) — but these operate on cross-sectional network snapshots rather than individual-level longitudinal sequences. To our knowledge, no study has applied persistent homology with structured null-model batteries to longitudinal socioeconomic trajectories.

Critical gaps remain: How does the topology of trajectory space relate to socioeconomic mobility regimes? Can we systematically test whether observed topological structure exceeds Markov baselines of varying complexity? And does the choice of test statistic — scalar summary versus diagram-level comparison — change the answer?

1.3 Contribution and Roadmap

We apply persistent homology to longitudinal life-course data and validate the resulting topology against a hierarchy of generative null models. The regime typology qualitatively resembles what sequence analysis finds in similar data (e.g., Bukodi and Goldthorpe, 2019); TDA adds multi-scale geometric characterisation and formal significance testing. A direct comparison with optimal matching yields ARI = 0.26 at matched $k = 7$ (§5.1).

The **Markov memory ladder** — five null models of increasing complexity (label shuffle, cohort shuffle, order shuffle, Markov order-1, Markov order-2) — decomposes the topological signal into contributions from state frequencies, temporal ordering, and memory length. We implement the ladder with two test statistics: total persistence (scalar) and Wasserstein diagram-level distances (geometric). The two give different answers for the Markov null. Total persistence says the topology is bounded by first-order dynamics; Wasserstein says the observed diagram is an outlier. This discrepancy — detectable only because we compared both statistics on the same null battery — has implications for TDA hypothesis testing more broadly.

We also analyse overlapping career-phase windows to quantify regime stickiness. Phase-order shuffle testing ($n = 500$) finds no evidence that phase ordering creates additional topology (H_0 $p = 0.98$, 95% CI [0.968, 0.992]; H_1 $p = 0.542$, 95% CI [0.498, 0.586]).

We find that H_0 topological structure (cluster separation) is order-dependent under total-persistence testing (order-shuffle $p < 0.005$) but not rejected under first- or second-order Markov nulls (Markov-1 $p = 1.000$, Markov-2 $p = 1.000$). However, upgrading to Wasserstein diagram-level distances reverses the Markov-1 conclusion: Markov-1 H_0 is rejected ($p = 0.002$), indicating that the connected-component structure of the trajectory manifold cannot be reproduced by a first-order Markov chain when diagrams are compared as geometric objects. Markov-2 is *not* rejected under Wasserstein ($p = 0.546$), localising the non-Markovian structure to second-order memory. Label shuffle serves as a negative control under both statistics ($p = 0.315$ total persistence, $p = 0.452$

Wasserstein). Order-shuffle H_0 is borderline under Wasserstein ($p = 0.058$), while H_1 remains non-significant under both statistics and all null types. These findings are stable across landmark counts from 2,500 to 8,000 (Appendix B) under total-persistence testing, replicate on BHPS-era trajectories under both total persistence and Wasserstein testing (§4.7), and are robust to embedding sensitivity testing across six configurations (§5.4). Stratified analysis reveals origin-dependent topology; all significant results survive Benjamini-Hochberg FDR correction. Overlapping 10-year windows confirm extreme regime stickiness (5.6% escape rate from disadvantaged regimes; 11.0% under non-overlapping windows).

2 Literature Review

2.1 Sequence Analysis in Social Science

The application of sequence analysis to life-course data originated with Abbott's (1995) proposal to adapt biological sequence techniques for sociological use. Optimal matching (OM) computes distances between state sequences using substitution and indel operations, analogous to edit distances in string matching. The resulting distance matrix is then used for cluster analysis (typically Ward's hierarchical clustering or partitioning around medoids) to identify trajectory "types".

OM has been extended in several directions: dynamic Hamming distances weight substitution costs by temporal proximity (Lesnard, 2010), multichannel OM enables joint analysis of employment and family states (Gauthier et al., 2010), and TraMineR provides a comprehensive software implementation (Studer and Ritschard, 2016). The resulting typologies span employment careers (Halpin and Chan, 1998), family formation (Elzinga and Liefbroer, 2007), and housing trajectories (Coulter et al., 2016).

Sequence analysis has specific limitations for the questions we pose. The dependence on substitution cost matrices introduces subjectivity: different cost specifications can yield different typologies from the same data (Robette and Thibault, 2008). The resulting distance measures are fundamentally pairwise — they compare sequences

but do not characterise the global geometry of the trajectory space. There is no natural notion of "closeness in trajectory space" that preserves topological invariants such as connected components, loops, or higher-dimensional voids. Significance testing is typically absent: cluster solutions are evaluated by silhouette scores or Calinski-Harabasz indices, but there is no formal test of whether the observed clustering exceeds what specific null models would produce.

2.2 Topological Data Analysis Foundations

Persistent homology (PH) provides a rigorous mathematical framework for detecting and quantifying topological features in data. Given a point cloud $X \subset \mathbb{R}^d$, PH constructs a filtration — a nested sequence of simplicial complexes $K_0 \subseteq K_1 \subseteq \cdots \subseteq K_m$ — by connecting points within distance ϵ and tracking which topological features appear (are "born") and disappear ("die") as ϵ increases (Edelsbrunner et al., 2002; Zomorodian and Carlsson, 2005).

The Vietoris–Rips (VR) complex, the most common choice for point cloud data, includes a k-simplex whenever all $\binom{k+1}{2}$ pairwise distances are below ϵ. For n points, the VR complex can have up to 2^n simplices, making direct computation infeasible for large datasets. Maxmin landmark selection (de Silva and Carlsson (2004)) addresses this by greedily selecting a subset of $L \ll n$ points that are maximally dispersed, constructing the VR complex on landmarks only. Ripser (Bauer, 2021) provides state-of-the-art computational efficiency, exploiting the apparent pairs optimisation to compute VR persistence through H_1 (and optionally H_2) on thousands of landmarks in minutes.

The output of PH is a persistence diagram (PD): a multiset of birth-death pairs (b_i, d_i) where feature i is born at filtration value b_i and dies at d_i. Features with large persistence $\ell_i = d_i - b_i$ are interpreted as genuine topological structure, while short-lived features near the diagonal are attributed to noise. In dimension 0 (H_0), features correspond to connected components: each merger of two components kills one H_0 feature. In dimension 1 (H_1), features correspond to loops or cycles: a loop forms at

birth and is filled in at death. Summary statistics include total persistence $\sum_i \ell_i$, maximum persistence $\max_i \ell_i$, and persistence entropy $-\sum_i \hat{\ell}_i \log \hat{\ell}_i$ where $\hat{\ell}_i = \ell_i / \sum_j \ell_j$ (Atienza et al., 2019).

A fundamental property of PH is stability: small perturbations in the input point cloud produce small changes in the persistence diagram, measured by the bottleneck or Wasserstein distance (Cohen-Steiner et al., 2007). Topological summaries are therefore robust to noise, sampling variation, and minor measurement errors — a property we rely on given the imputation and harmonisation challenges in survey data.

For comparing persistence diagrams across groups, the Wasserstein distance $W_p(D_1, D_2)$ measures the cost of the optimal transport between two diagrams, providing a principled metric for testing whether subpopulations have different topological structure (Turner et al., 2014). As we demonstrate in this paper, the choice between scalar summaries (total persistence) and diagram-level comparisons (Wasserstein distance) can change substantive conclusions — a point that has methodological implications beyond our specific application.

2.3 Mobility Regimes and Their Topological Signatures

The economics of poverty traps posits that disadvantaged states can be self-reinforcing through multiple equilibria (Azariadis and Stachurski, 2005). When returns to human capital investment are nonlinear — or when credit constraints, neighbourhood effects, or discrimination create threshold dynamics — individuals may become trapped in low-productivity equilibria from which escape requires large, discrete shocks rather than marginal improvements (Barrett and Carter, 2013).

Life-course researchers have identified analogous patterns in longitudinal data. Vandecasteele (2010) distinguished between stable, mobile, and churning poverty trajectories using Belgian panel data. Jenkins (2011) documented extensive income churning in the UK, where many individuals cycle between poverty and near-poverty without achieving durable exits. Savage et al. (2013) used the Great British Class Survey to identify seven social classes, some characterised by cultural and social capital deficits

that reinforce economic disadvantage.

The connection to topology runs through homological dimension. If mobility regimes exist as distinct clusters in trajectory space, they will appear as persistent H_0 components (connected components). The persistence values of H_0 features quantify the separation between regime clusters: large persistence indicates well-separated regimes that merge only at large filtration radii.

H_1 features (loops) have a distinct geometric meaning. Persistent H_1 loops correspond to cyclical oscillation patterns in the embedding space — subsets of trajectories that trace closed paths through state space, returning to similar configurations after a sequence of transitions. This is geometrically distinct from the absorbing-state dynamics that characterise poverty traps in the economic literature (Azariadis and Stachurski, 2005). Poverty traps as absorbing states would manifest as dense H_0 clusters — high-density regions in the point cloud from which few trajectories escape — rather than as H_1 loops. The distinction matters: H_1 tests whether the trajectory manifold contains persistent cyclical structure, not whether disadvantaged states are absorbing.

In our data, H_1 is not rejected under any null model, including the first-order Markov null, regardless of which test statistic is used (§4.3). This bounds the complexity of cyclical dynamics: the trajectory manifold lacks persistent cyclical structure beyond what first-order Markov dynamics produce — ruling out non-Markovian topological trapping cycles while directing attention toward H_0 cluster absorptivity and escape rates (§4.4.2) as the appropriate lens for poverty-trap dynamics.

The mapping between topological features and substantive social phenomena must respect the geometric meaning of each homological dimension; a topological analysis complements both sequence typologies and Markov transition models, but only when this mapping is drawn carefully.

2.4 Statistical Null Models for TDA

The statistical significance of topological features requires comparison to appropriate null distributions. Robinson and Turner (2017) proposed permutation tests for PH,

comparing observed total persistence or Betti numbers to distributions generated by permuting group labels. Stolz et al. (2017) applied PH to time-dependent functional networks, using surrogate data methods as nulls. These contributions establish the general framework; our contribution is to adapt and extend it with a structured hierarchy of null models tailored to sequential socioeconomic data.

For life-course trajectories, the relevant null models must be carefully chosen to test specific hypotheses. A label shuffle null (permuting state labels globally) tests whether the specific identity of states matters; an order shuffle (permuting each trajectory's state sequence) tests whether temporal ordering matters, preserving marginal state frequencies; and a Markov-k null (simulating from an estimated k-th order Markov chain) tests whether topology exceeds what k-step memory produces.

Higher-order Markov chains as generative models have a long history in sociology (Singer and Spilerman, 1976; Tong, 1990). Our contribution is to arrange these null models into a systematic "memory ladder" — a sequence of increasingly structured null hypotheses that progressively preserves more of the observed data's structure. When the observed topology first becomes consistent with a Markov-k null, we can conclude that the topological complexity of the trajectory space is well captured by k-step memory. Critically, the answer depends on how "consistent" is measured. As we show, total persistence (a scalar) and Wasserstein distance (a diagram-level metric) give different answers for the Markov null — the scalar test lacks sensitivity to structural differences in the configuration of persistence features that the Wasserstein test detects.

3 Data and Methodology

3.1 Data: BHPS and Understanding Society

Our data combine two longitudinal household surveys: the British Household Panel Study (BHPS, waves 1–18, 1991–2008) and Understanding Society (USoc, waves 1–14, 2009–2023). BHPS began as a nationally representative sample of approximately

5,500 households in Great Britain, with annual interviews tracking employment, income, and demographic characteristics. Understanding Society succeeded BHPS with a larger initial sample of approximately 40,000 households and incorporated the surviving BHPS sample as a sub-sample from wave 2 (2010–2011), enabling longitudinal tracking across both surveys.

Our primary analysis uses the USoc-era panel. From the combined dataset of over 118,000 respondents, we select individuals with at least 10 consecutive years of valid employment-income observations within the USoc observation window, yielding 27,280 trajectories. The mean trajectory length is 12.9 years (median 13, range 10–14), with start years spanning 2009–2013 and end years 2018–2022.

As cross-era validation, we separately construct BHPS-era trajectories (1991–2008) by linking BHPS person identifiers (`{wave}pid`) to USoc identifiers (`pidp`) via the cross-wave identifier file (`xwavedat.tab`). BHPS raw monthly net household income (`fihhmn`) is equivalised by dividing by the modified OECD equivalence scale (`eq_moecd`) from the household response file, matching the equivalisation approach used for USoc's `fihhmnnet1_dv`. This yields 8,509 BHPS-era trajectories (mean length 14.5 years). We also identify 8,459 spanning trajectories observed in both surveys. Cross-era results are reported in §4.7.

Each person-year is assigned to one of nine states formed by crossing employment status with threshold-based income category. Employment status is derived from the `jbstat` variable: employed (E) includes full-time and part-time employment and self-employment; unemployed (U) includes ILO-definition unemployment; inactive (I) includes retirement, full-time education, long-term sickness, and caring responsibilities. Income bands are defined annually relative to the contemporary equivalised household income median: low (L, below 60% of median), mid (M, 60%–120% of median), and high (H, above 120% of median). The L/M/H classification uses fixed boundaries, not population-quantile cut-points; the resulting category sizes are unequal and vary across years. Sensitivity to alternative threshold choices (e.g., 50%/150% or population terciles) is not examined here.

Gaps of one or two years in otherwise continuous trajectories are forward-filled from the last observed state, affecting 4.9% of person-years. We verify that imputation rates are balanced across regimes and do not drive topological results (Supplementary Material SA).

For stratification analysis, we use three covariates: sex (Female n = 14,402; Male n = 11,243; 1,635 with missing sex excluded from gender stratification); parental NS-SEC (Professional/Managerial n = 2,067; Intermediate n = 6,777; Routine/Manual n = 7,779; 10,657 missing); and birth cohort decade (1930s n = 1,642 through 1990s n = 1,364, with seven cohort groups).

Parental NS-SEC is missing for 39.1% of the sample (10,657 of 27,280). This missingness is non-random: a chi-squared test of regime membership by NS-SEC availability yields $\chi^2 = 147.74$ ($df = 6$, $p < 10^{-29}$), and logistic regression confirms that regime membership, birth cohort, and trajectory characteristics all predict NS-SEC availability. All NS-SEC-stratified results (§4.6) should be interpreted with this caveat.

3.2 Trajectory Embedding

To embed trajectories in a continuous metric space suitable for TDA, we compute n-gram frequency vectors. For each trajectory of length T, we calculate:

- **Unigrams** (9 dimensions): Relative frequency of each state $s \in \{EL, EM, EH, UL, UM, UH, IL, IM, IH\}$ in the trajectory.
- **Bigrams** (81 dimensions): Relative frequency of each consecutive state pair (s_t, s_{t+1}), capturing transition structure.

The concatenated 90-dimensional term-frequency (TF) vector encodes both the composition and the sequential structure of each trajectory. We standardise using a fitted `StandardScaler` and project to 20 dimensions via Principal Component Analysis (PCA), retaining the top 20 components. The resulting point cloud of 27,280 points in $\mathbb{R}^{20}$ preserves both global cluster structure and local sequence variation.

As a sensitivity check, we compute UMAP embeddings with 16 dimensions and compare GMM cluster assignments to those from PCA-20D. The adjusted Rand index (ARI) between the two clusterings is 0.33 at matched $k = 7$, indicating that while three broad macro-types (secure employment, inactive/marginal, churning/mixed) are stable across embedding methods, finer regime boundaries are embedding-dependent. UMAP and PCA capture different aspects of local versus global geometry (McInnes et al., 2018), which explains the divergence at fine scales. We report PCA-20D as the primary embedding and UMAP-16D as a supplementary sensitivity analysis (Figure S1).

To verify that substantive conclusions are robust to embedding choice, we re-run the order-shuffle null on the UMAP-16D embedding ($n = 100$). With GMM fixed at $k = 7$, UMAP yields ARI = 0.33 against PCA regime assignments. On the UMAP-16D point cloud with 5,000 maxmin landmarks, the order-shuffle null rejects H_0 at $p < 0.001$ (observed total persistence 3,203.3, null mean 2,372.9 $\pm$ 323.3) while H_1 is non-significant ($p = 1.00$), reproducing the PCA pattern. Full Markov-1 replication on UMAP is omitted due to computational cost; given the close agreement in regime structure and order-shuffle results, we treat PCA as the primary embedding and UMAP as a robustness check.

3.3 Persistent Homology

We compute Vietoris–Rips persistent homology on the PCA-20D point cloud using the following pipeline:

1. **Maxmin landmark selection**: From the 27,280 embedded trajectories, we select L landmarks using the greedy furthest-point (maxmin) algorithm (de Silva and Carlsson (2004)). This selects an initial seed point and iteratively adds the point maximally distant from the current landmark set, producing a well-dispersed subsample that preserves the geometric structure of the full cloud. For total-persistence tests, $L = 5{,}000$ (approximately 18% of the point cloud); for Wasserstein tests, $L = 2{,}000$ (approximately 7%). We verify robustness to this choice by recomputing PH at $L = 2{,}500$ and

$L = 8{,}000$ (Appendix B).

2. **VR complex construction and PH computation**: We construct the Vietoris–Rips filtration on the landmarks and compute persistence through H_1 using Ripser (Bauer, 2021), with the filtration threshold set to the 75th percentile of pairwise distances.

3. **Persistence diagrams and summaries**: The output comprises H_0 (connected components) and H_1 (loops) persistence diagrams. For each dimension, we compute total persistence $\sum_i (d_i - b_i)$, maximum persistence, feature count, and persistence entropy.

H_0 features correspond to the merging of trajectory clusters as the filtration radius increases: long-lived H_0 features indicate well-separated regimes. H_1 features correspond to loops in the trajectory space: persistent loops may indicate cyclical trapping where trajectories circulate among a subset of states without converging to a single attractor.

3.4 Null Model Battery: The Markov Memory Ladder

We introduce a battery of five null models arranged in order of increasing structural fidelity to the observed data. For each null model, we generate surrogate trajectory datasets and process them through the same pipeline: re-embed using the n-gram + PCA pipeline with PCA loadings fixed from the observed data (i.e., surrogate n-gram vectors are projected onto the observed PCA axes, not recomputed). Maxmin landmarks are re-selected on each surrogate to avoid coupling landmark geometry to the observed data. We then recompute PH and compare to the observed data.

Null 1 — Label shuffle (n = 200): Permute state labels globally across all person-years, destroying all individual-level structure.

Null 2 — Cohort shuffle (n = 200): Permute trajectories within birth-cohort bins, testing whether cross-cohort differences drive the topological signal.

Null 3 — Order shuffle (n = 100): For each individual, randomly permute the temporal order of their state sequence, preserving per-person state frequencies but destroying all transition and temporal structure.

Null 4 — Markov order-1 ($n = 100$): Estimate a first-order Markov transition matrix from the observed data via maximum likelihood and simulate surrogate trajectories of the same lengths, conditional on each trajectory's observed starting state.

Null 5 — Markov order-2 ($n = 200$): Estimate a second-order Markov transition matrix (conditioning on the previous two states) via maximum likelihood, with Laplace smoothing (adding pseudocounts of 1 to all 729 cells). Surrogates are simulated conditional on the observed first two states of each trajectory.

3.4.1 Test statistics

We implement two test statistics to compare observed and null persistence:

Total persistence (scalar): $\sum_i (d_i - b_i)$ for each homological dimension, separately for H_0 and H_1. The p-value is the proportion of null realisations with total persistence as extreme as or more extreme than the observed value. This is the standard approach in TDA hypothesis testing (Robinson and Turner, 2017).

Wasserstein diagram distance (diagram-level): For each null realisation j, we compute $W_1(D_{\text{obs}}, D_{\text{null}_j})$ — the 1-Wasserstein distance between the observed persistence diagram and the null diagram, using the `gudhi.wasserstein` implementation backed by the POT (Python Optimal Transport) library. We summarise these as the *mean* observed-null distance $\bar{W} = \frac{1}{n} \sum_j W_1(D_{\text{obs}}, D_{\text{null}_j})$; using the mean rather than individual distances averages over sampling noise in the null realisations and produces a stable scalar test statistic. We also compute a null-null baseline: $W_1(D_{\text{null}_i}, D_{\text{null}_j})$ for 500 sampled pairs of null diagrams, which characterises the typical spread within the null distribution and accounts for its variance. The p-value is the proportion of null-null distances that equal or exceed $\bar{W}$. This approach tests whether the observed diagram is more distant from the null distribution than null diagrams are from each other — i.e., whether the observed diagram is an outlier relative to the null cloud in Wasserstein space.

The Wasserstein test is more sensitive than total persistence because it compares the full configuration of birth-death pairs rather than collapsing them to a scalar. Two

diagrams can have identical total persistence but very different geometric structure: for instance, many short-lived features versus few long-lived ones. The Wasserstein distance detects such shape differences; total persistence does not.

With $n = 100$ for the order-shuffle and Markov-1 nulls, we have resolution to approximately $p = 0.01$ for one-sided tests. For BHPS-era validation (§4.7), null tests use $n = 500$ permutations.

3.5 Regime Discovery

We identify mobility regimes by fitting Gaussian mixture models (GMMs) to the PCA-20D embedding, using the Bayesian Information Criterion (BIC) for model selection across $k = 3, 4, \ldots, 15$ components. The BIC-optimal model has $k = 7$ components.

H_0 informs but does not replicate GMM: all H_0 features are born at filtration radius 0 (each landmark starts as a singleton); the 44 features with persistence > 10 merge gradually (7th/8th most persistent ratio = 1.03×), reflecting single-linkage chaining in 20-dimensional Vietoris–Rips complexes on 5,000 landmarks. The merge tree yields one dominant connected component (99.98% of trajectories) plus 6 singleton outliers — no balanced clusters matching GMM density modes. We use GMM for regime labels and H_0 for connectivity validation (§4.4.1, Appendix A).

Regime profiles are constructed by mapping GMM-assigned labels back to the original state sequences and computing, for each regime: employment rate, unemployment rate, inactivity rate, income distribution (fraction in low/mid/high), and a stability index (fraction of trajectory years spent in the dominant state).

To verify that the $k = 7$ solution is not an artefact of a particular sample, we perform a bootstrap stability analysis: 200 resamples of the 27,280 trajectory embeddings with replacement, each fitted with GMM $k = 7$, yielding ARI distribution with mean 0.646 ± 0.086 (95% CI: [0.461, 0.795]).

3.6 Overlapping Windows and Regime Persistence

To assess regime persistence over the life course, we construct overlapping career-phase windows using a sliding window of 10 years with a 5-year step. For each individual, this produces a sequence of overlapping windows, each embedded and assigned to a regime using the same PCA-GMM model fitted on full trajectories. This yields 54,560 windows from 27,280 individuals (mean 2.0 windows per person).

We construct a 7×7 regime transition matrix from consecutive windows and compute escape probabilities: the fraction of individuals starting in disadvantaged regimes (Low-Income Churn or Inactive Low) who reach an advantaged regime (Secure EH or Employed Mid) within a given number of windows.

We also compute PH on the phase-level embedding cloud and test a phase-order shuffle null that preserves each individual's multiset of windows but randomises their temporal order, testing whether the life-course ordering of career phases adds topological structure beyond the composition of phases. We run $n = 500$ permutations with 3,000 maxmin landmarks and report p-values with 95% confidence intervals.

3.7 Sequence Analysis Baseline

To contextualise our TDA results, we compute a standard sequence analysis baseline using optimal matching (OM) with dynamic Hamming distances (Lesnard, 2010). We compute pairwise distances between all 27,280 trajectories, apply Ward's hierarchical clustering, and select k via the average silhouette width (Studer and Ritschard, 2016).

The OM-optimal solution selects $k = 2$ clusters (silhouette = 0.223), producing a coarse division between predominantly inactive trajectories (40.2%) and predominantly employed trajectories (59.8%). At $k = 7$, OM achieves silhouette = 0.215. The ARI between OM $k = 7$ clusters and GMM $k = 7$ regimes is 0.26, indicating overlapping but non-identical regime boundaries. Regime-level profile correlations range from 0.97 (high-employment, which both methods identify cleanly) down to 0.02 (mixed-churn, where OM and TDA draw boundaries differently).

The moderate ARI confirms that the broad structure of UK trajectory space — se-

cure employment versus inactivity versus churning — is recoverable by either method, while finer distinctions depend on the analytic approach. The primary added value of TDA lies not in the regime typology itself but in the formal null testing infrastructure.

4 Results

4.1 Descriptive Statistics

Our sample comprises 27,280 individuals with trajectories averaging 12.9 years (range 10–14, median 13). The nine-state distribution reflects a labour market with dominant employment: EH (high-income employment) is the most common dominant state (42.1% of individuals), followed by IL (inactive-low, 13.7%), EM (employed-mid, 12.5%), IM (inactive-mid, 12.1%), and IH (inactive-high, 12.0%). Unemployment is comparatively rare as a dominant state: UL (1.1%), UM (0.5%), and UH (0.4%). The low-income employment state EL dominates 5.7% of trajectories.

Birth cohorts span seven decades, from the 1930s (n = 1,642) through the 1990s (n = 1,364), with the largest cohorts in the 1960s (n = 5,579) and 1950s (n = 4,843). The gender split is 56% female, 44% male among those with non-missing sex. Forward-fill imputation affected 4.9% of person-years, distributed proportionally across regimes.

Table 1 summarises sample characteristics by cohort decade.

4.2 Topological Structure of the Trajectory Space

Figure 2 shows the UMAP projection of the PCA-20D embedding; Figure 3 shows the resulting persistence diagrams.

Persistent homology on the PCA-20D embedding with 5,000 maxmin landmarks reveals substantial topological structure in both dimensions.

H_0 **(connected components):** 5,000 features (4,999 finite, 1 infinite — the final connected component), with total persistence 20,411.1 and maximum persistence 15.81. The dominance of a single long-lived component (persistence 15.81, compared to the

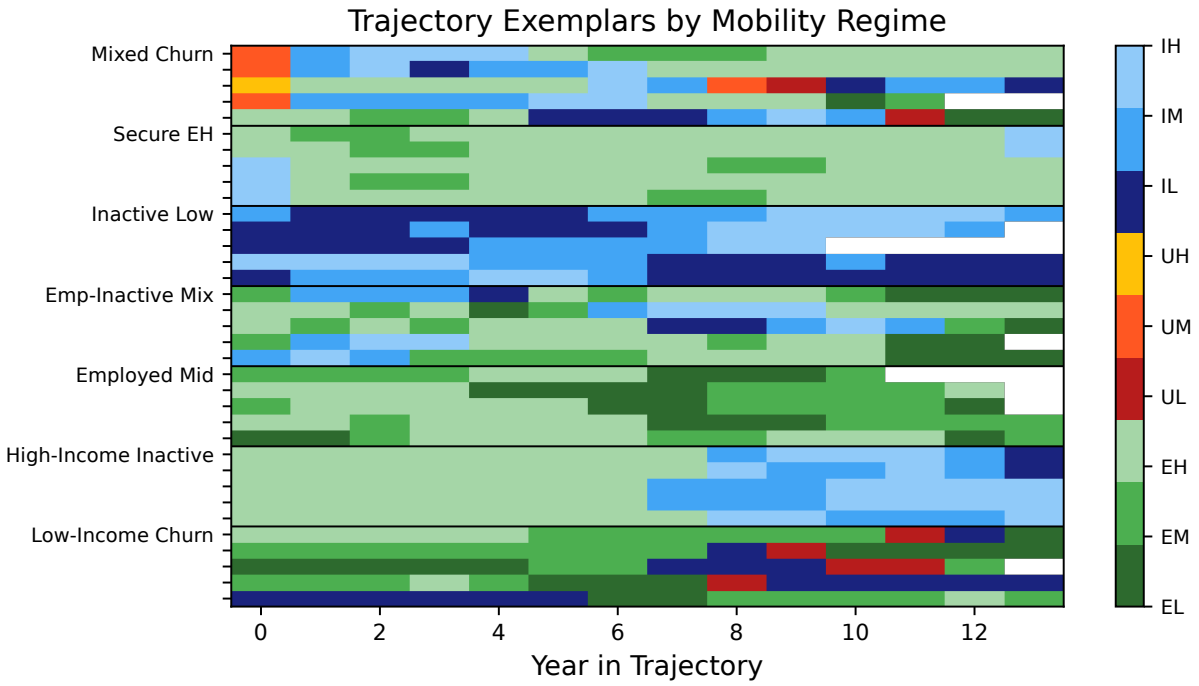

Figure 1: Exemplar trajectories from each of the seven mobility regimes. Each row represents one individual; colours encode employment-income state.

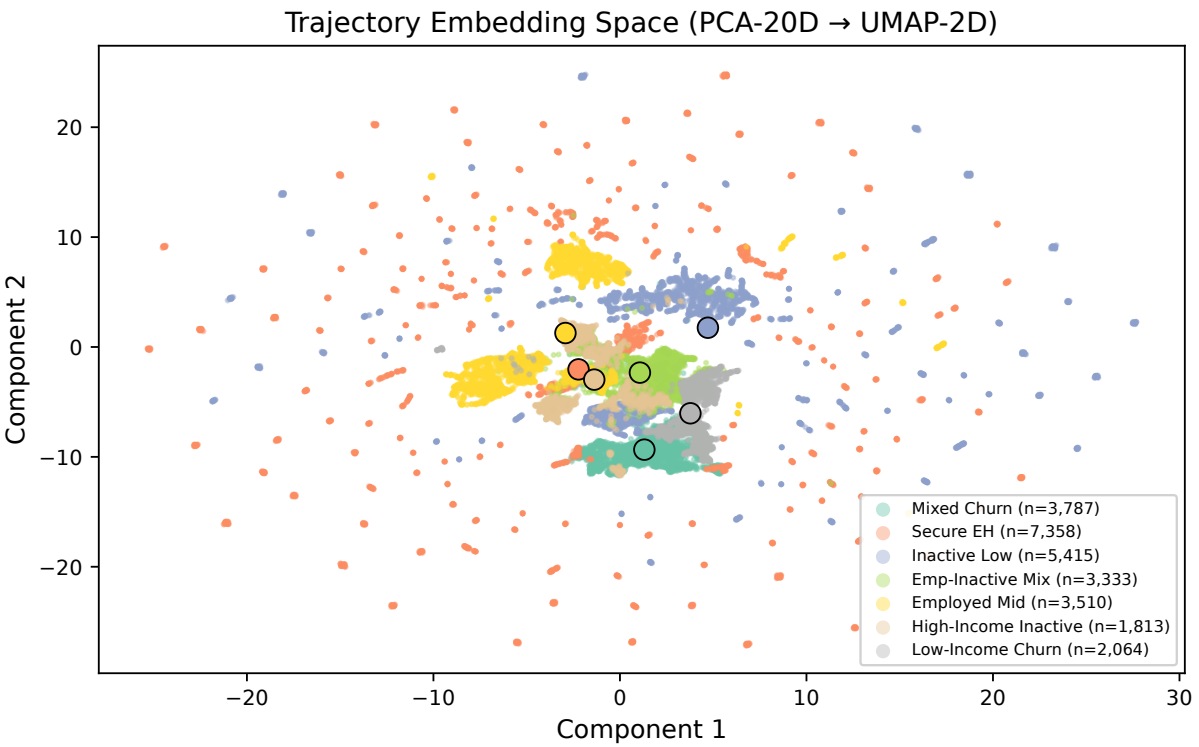

Figure 2: Trajectory embedding space (bigram PCA-20D reduced to UMAP-2D). Points are individual career trajectories, coloured by GMM regime.

mean of 4.08) indicates that most trajectory clusters ultimately merge at large filtration values but are well separated at smaller scales. H_0 reflects unimodal connectivity plus outliers: 99.98% of trajectories belong to a single giant connected component, with 6 singleton landmarks merging late — consistent with single-linkage chaining in high-dimensional Vietoris–Rips complexes. The distribution of H_0 persistence values has a heavy right tail, but the top features merge gradually (7th/8th most persistent ratio = 1.03×), indicating no sharp topological gap between clusters.

H_1 **(loops):** 5,962 features (5,961 finite, 1 infinite), with total persistence 2,224.7 and maximum persistence 3.21. Loops are numerous but not deeply persistent: the maximum H_1 persistence (3.21) is a fifth of the maximum H_0 persistence (15.81).

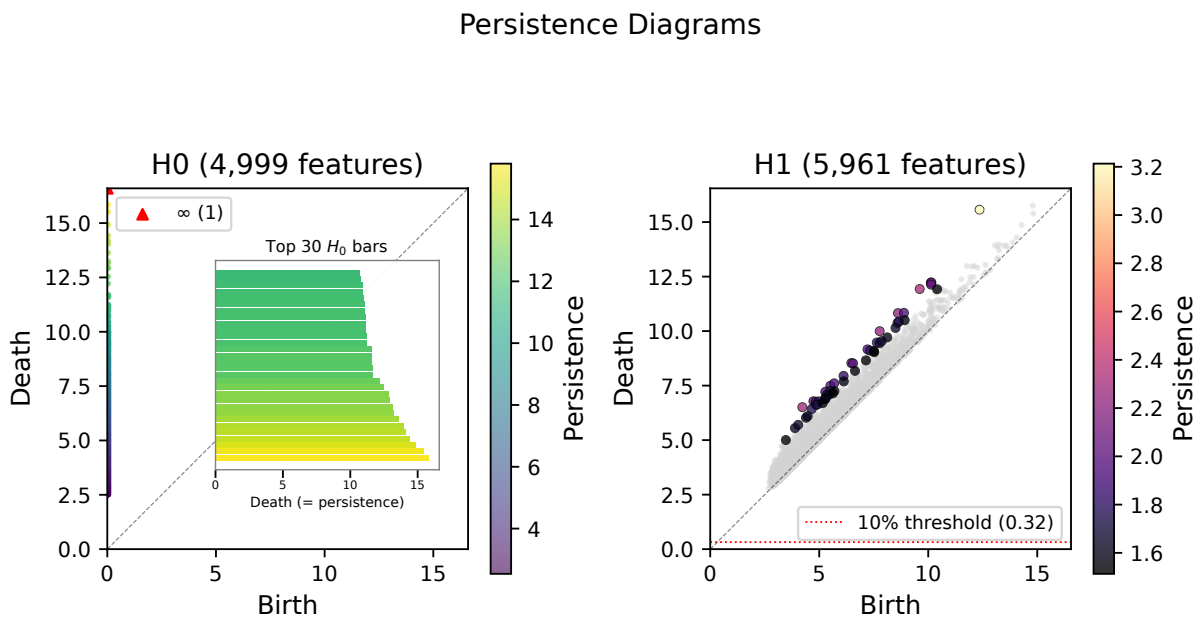

Figure 3: Persistence diagrams for H_0 (connected components, left) and H_1 (loops, right). Points far from the diagonal represent long-lived topological features. In Vietoris–Rips H_0, all components are born at filtration radius 0, so every point lies on the y-axis; the inset barcode of the 30 longest-lived features makes the gap structure readable (persistence equals death time).

4.3 Null Model Validation: The Markov Memory Ladder

4.3.1 Total-persistence tests

The five null models in the Markov memory ladder yield the following results for total persistence at $L = 5{,}000$ landmarks (Table 2):

Table 1: Markov Memory Ladder — Total Persistence Results ($L = 5{,}000$ landmarks).

Null Model	n	H_0 obs	H_0 null ($\mu \pm \sigma$)	H_0 p	H_1 obs	H_1 null ($\mu \pm \sigma$)	H_1 p
Label shuffle	200	20,411.1	20,405.4 ± 10.3	0.315	2,224.7	2,234.6 ± 13.3	0.710
Cohort shuffle	200	20,411.1	20,407.8 ± 9.6	0.405	2,224.7	2,234.5 ± 12.7	0.725
Order shuffle	100	20,411.1	19,364.2 ± 73.8	**< 0.005**	2,224.7	2,156.7 ± 30.8	**0.020**
Markov-1	100	20,411.1	21,138.3 ± 196.8	1.000	2,224.7	2,381.3 ± 63.8	1.000
Markov-2	200	20,411.1	21,323.8 ± 174.2	1.000	2,224.7	2,253.3 ± 47.1	0.735

Under total persistence, the ladder tells a clean but, as we show below, incomplete story. Temporal ordering shapes the global geometry of the point cloud (order-shuffle H_0 $p < 0.005$). Markov surrogates generate *more* total persistence than observed (null mean H_0 = 21,138.3 vs observed 20,411.1), yielding $p = 1.000$ — the observed value sits below the null distribution. Label and cohort shuffles serve as negative controls (both non-significant, as expected). Order-shuffle H_1 is significant at $L = 5{,}000$ ($p = 0.020$) but landmark-sensitive (non-significant at $L = 2{,}500$ and $L = 8{,}000$; Appendix B).

Taken alone, these results would suggest that first-order Markov dynamics fully account for the topology of the trajectory point cloud. The scalar summary indicates that Markov surrogates produce as much or more total persistence as the real data. The Wasserstein results below revise this conclusion.

4.3.2 Wasserstein diagram-level tests

We re-run the full Markov memory ladder using Wasserstein diagram distances as the test statistic, on $L = 2{,}000$ landmarks with $n = 100$ permutations per null type. For each null realisation, we compute $W_1(D_{\text{obs}}, D_{\text{null}})$ per homological dimension, and compare to a null-null baseline of 500 sampled pairs.

Label shuffle serves as a negative control: the observed diagram is indistinguishable from shuffled diagrams (H_0 $p = 0.452$, H_1 $p = 0.538$), confirming that the specific identity of states does not drive diagram geometry when all temporal structure is preserved. Cohort shuffle Wasserstein results are omitted because the checkpoint metadata does not store birth-cohort assignments in a format amenable to the Wasserstein testing pipeline; the total-persistence cohort-shuffle result ($p = 0.405$) establishes that

Table 2: Markov Memory Ladder — Wasserstein Distances ($L = 2{,}000$ landmarks, $n = 100$).

Null Model	Dim	W(obs↔null) $\mu \pm \sigma$	W(null↔null) $\mu \pm \sigma$	p	Sig?
Label shuffle	H_0	0.84 ± 0.16	0.83 ± 0.18	0.452	no
Label shuffle	H_1	134.81 ± 1.67	134.64 ± 6.25	0.538	no
Order shuffle	H_0	11.22 ± 1.52	6.62 ± 2.37	0.058	borderline
Order shuffle	H_1	139.69 ± 3.07	144.66 ± 4.61	0.854	no
Markov-1	H_0	11.42 ± 2.15	5.99 ± 1.79	**0.002**	**yes**
Markov-1	H_1	126.85 ± 2.92	121.39 ± 3.95	0.086	no
Markov-2	H_0	6.62 ± 1.72	7.19 ± 2.09	0.546	no
Markov-2	H_1	129.71 ± 2.75	123.21 ± 4.71	0.078	no

cross-cohort differences do not drive the topological signal.

The Wasserstein results change the Markov conclusion. The observed H_0 persistence diagram is significantly different from Markov-1 null diagrams ($p = 0.002$): the mean Wasserstein distance between observed and null diagrams (11.42 ± 2.15) is nearly double the mean distance between pairs of null diagrams (5.99 ± 1.79). The observed diagram is an outlier relative to the Markov null cloud in diagram space, even though it has less total persistence. The Markov process generates *more* features overall but in a different geometric configuration — different birth-death pairs, different persistence structure — than the real data.

Order-shuffle H_0 is borderline under Wasserstein ($p = 0.058$), weaker than the clear rejection under total persistence ($p < 0.005$). This implies that temporal ordering creates aggregate persistence differences (total persistence is higher in ordered data) but the diagram-level structure is only marginally different from the order-shuffled baseline. The effect of temporal ordering is real but operates more through the sum of feature lifetimes than through specific birth-death configurations.

H_1 is non-significant under both null types, consistent with the total-persistence results. Markov-1 H_1 is borderline ($p = 0.086$) — closer to significance under Wasserstein than under total persistence ($p = 1.000$) — suggesting that diagram-level H_1 differences may emerge with larger permutation counts, though we do not pursue this here.

Markov-2 resolves the Wasserstein discrepancy. Upgrading from first- to second-order Markov dynamics eliminates the diagram-level rejection: Markov-2 H_0 is not rejected under Wasserstein ($p = 0.546$), with the mean obs-null distance (6.62 ± 1.72)

falling well within the null-null distribution (7.19 $\pm$ 2.09). Markov-2 H_1 is also non-significant (p = 0.078), though closer to the boundary than Markov-1 H_1. This result localises the non-Markovian structure detected by the Wasserstein test: the diagram-level differences between observed and Markov-1 surrogates reflect second-order memory — dependence on the state two steps back — rather than higher-order or non-Markovian effects that no finite-memory model could capture. The Markov memory ladder thus yields a sharp conclusion: total persistence is insensitive to first-order violations (Markov-1 p = 1.000), Wasserstein detects them (p = 0.002), and second-order Markov dynamics suffice to reproduce the observed diagram geometry (p = 0.546).

4.3.3 The discrepancy between test statistics

The discrepancy between total persistence and Wasserstein distance for the Markov-1 H_0 test is the structurally interesting finding of this analysis. Total persistence says: the observed data has *less* aggregate persistence than the Markov null, so the null is not rejected. Wasserstein says: the observed data has a *different shape* of persistence diagram than the Markov null, and this difference is larger than the variation among null diagrams, so the null *is* rejected. Both are correct statements about different aspects of the comparison. The Markov process generates a more diffuse trajectory cloud (more total persistence, more features) but the specific pattern of which components form and when they merge differs from the observed data.

We interpret this as follows: the connected-component structure of the trajectory manifold — which clusters form, at what filtration scale, and in what hierarchical relationship — carries information about the data-generating process that a first-order Markov chain cannot reproduce. The Markov chain gets the total amount of persistence roughly right (in fact, overshoots it) but distributes it differently across birth-death pairs. The trajectory data has regime-specific substructure that the globally estimated Markov transition matrix smears out.

This finding has methodological implications beyond our specific application. Scalar persistence summaries are standard in TDA hypothesis testing, but they can produce

false negatives when the null generates as much or more aggregate persistence while differing in diagram shape. Wasserstein diagram comparisons should be considered alongside total-persistence tests in any TDA null-testing framework.

Intra-regime compactness confirms temporal order contributes to regime differentiation. We compute mean within-regime pairwise distances in the PCA-20D embedding under 500 order-shuffled re-embeddings, with PCA loadings frozen from the observed data. All seven regimes show significantly tighter compactness in observed versus order-shuffled embeddings (z-scores ranging from -30.5 to -67.9, all $p < 0.001$). Regime 0 (Mixed Churn, observed mean distance 14.68, null 19.12, z = -55.0) and Regime 6 (Low-Income Churn, observed 9.52, null 14.58, z = -67.9) show the largest compactness ratios, temporal ordering is most informative for the churning regimes, where transition sequences rather than state composition define the trajectory type.

Bootstrap null stability. To verify that null-model conclusions are not artefacts of a particular GMM partition, we conduct a bootstrap stability analysis: 10 bootstrap resamples of the trajectory embeddings, each re-fitted with GMM ($k = 7$), with order-shuffle and Markov-1 null tests (50 permutations each, 500 landmarks). Order-shuffle H_0 is significant in all 10/10 draws (p_{mean} = 0.016, range [0.00, 0.04]). Markov-1 H_0 is significant in all 10/10 draws (p_{mean} = 0.000). No conclusion flips are observed across any bootstrap draw.

4.4 Seven Mobility Regimes

BIC model selection identifies $k = 7$ as optimal (Figure 2 shows the UMAP projection coloured by regime). The seven regimes, ordered by size, are:

Regime 1 — Secure High-Employment (n = 7,358, 27.0%): Dominated by high-income employment (EH). Employment rate 92.3%, inactivity rate 7.4%, unemployment near zero (0.3%). High-income rate 85.5%. Stability index 0.779, the highest of all regimes.

Regime 2 — Inactive Low (n = 5,415, 19.9%): Entirely inactive (inactivity rate 100%, zero employment or unemployment). Income is mixed (low 38.7%, mid 33.1%, high

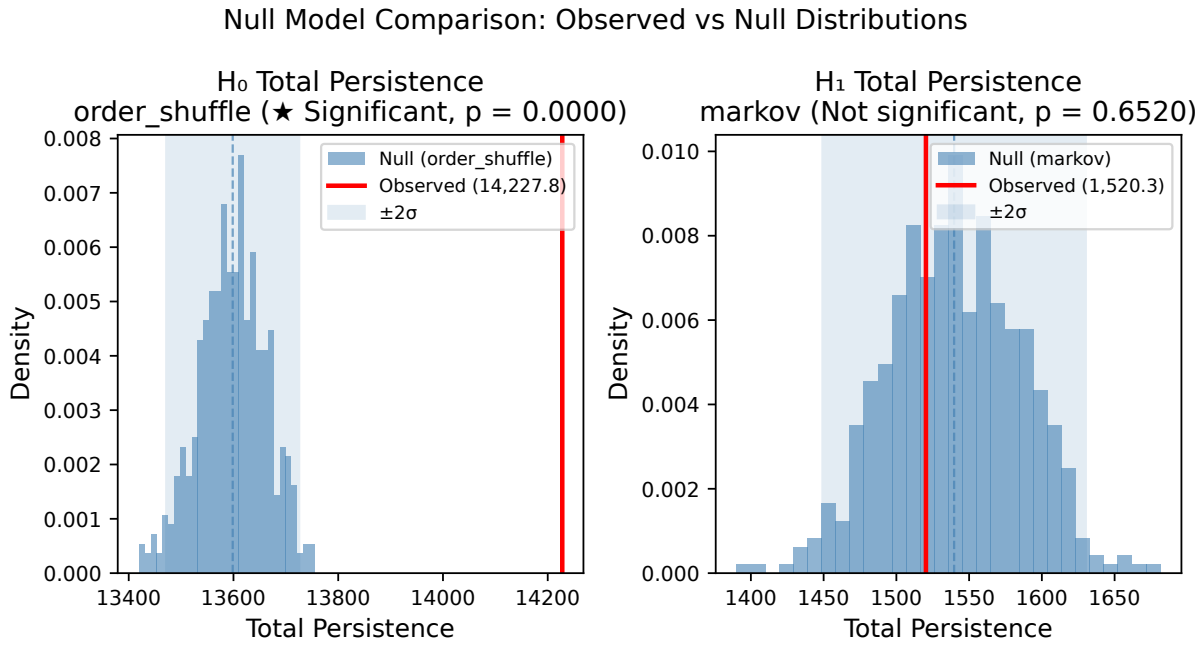

Figure 4: Observed total persistence versus order-shuffle null (left: H_0; right: H_1). Red line = observed; histogram = null distribution.

Null Model Validation Summary

Null Model	H₀ Obs	H₀ Null (μ ± σ)	H₀ z	H₀ p	H₀ Sig	H₁ Obs	H₁ Null (μ ± σ)	H₁ z	H₁ p	H₁ Sig
Label Shuffle	14,227.8	14,232.6 ± 9.7	-0.49	0.6900		1,520.3	1,524.3 ± 15.5	-0.26	0.6300	
Cohort Shuffle	14,227.8	14,230.9 ± 9.9	-0.31	0.6600		1,520.3	1,521.0 ± 16.2	-0.04	0.5400	
Order Shuffle	14,227.8	13,598.5 ± 63.4	9.92	**0.0000**	★	1,520.3	1,503.7 ± 26.7	0.62	0.2500	
Markov (order 1)	14,227.8	14,111.2 ± 118.1	0.99	0.1480		1,520.3	1,539.7 ± 45.2	-0.43	0.6520	
Markov (order 2)	14,227.8	14,077.4 ± 121.3	1.24	0.1080		1,520.3	1,511.9 ± 37.7	0.22	0.4340	

Figure 5: Null model validation summary across the Markov memory ladder. Significant results ($p < 0.05$) are shaded green.

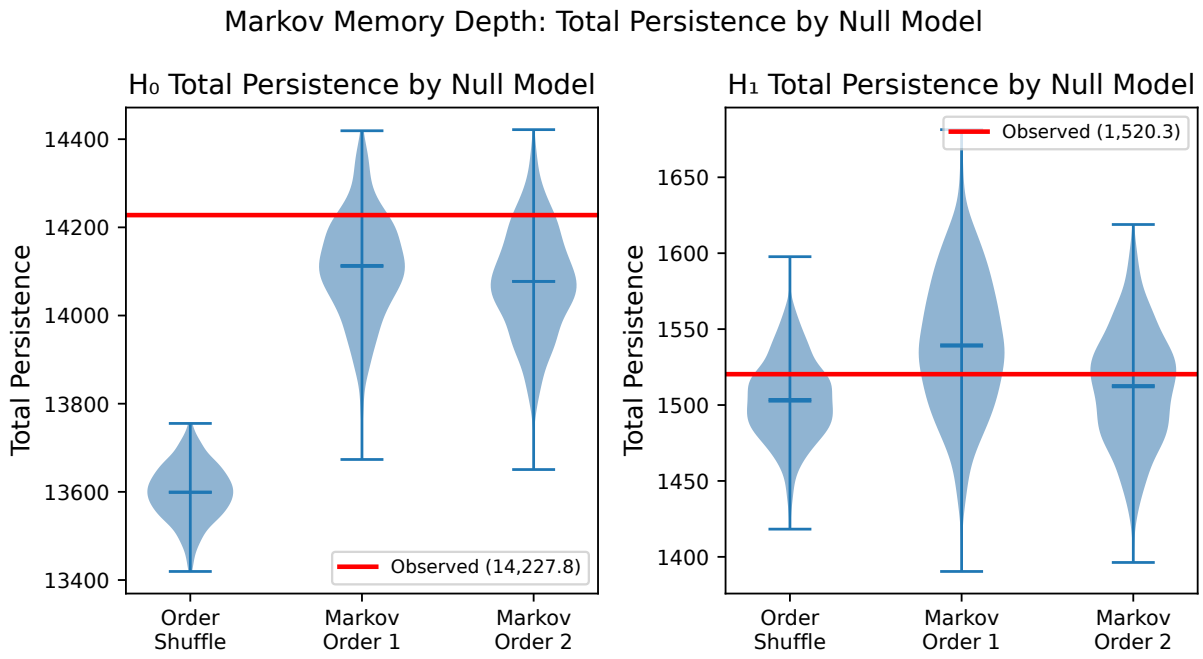

Figure 6: Total persistence distributions for order-shuffle, Markov order-1, and Markov order-2 null models. Red line = observed value.

28.2%; notably, 28.2% are high-income, reflecting pensioners with accumulated assets) — the regime groups retirees with pensions alongside carers and those with long-term illness. Stability 0.387. This regime is a key disadvantaged state, though we note that it likely mixes lifecycle inactivity (retirement) with inequality-driven inactivity (disability, long-term sickness, caring). The near-absorbing nature of this regime (§4.4.2) may partly reflect the irreversibility of retirement rather than a "trap" in an inequality sense. We address this via age-stratified analysis in §4.4.2.

Regime 0 — Mixed Churn (n = 3,787, 13.9%): Mixed employment (41.6%), unemployment (20.3%), and inactivity (38.1%). Income concentrated in mid (34.1%) and high (42.2%). Low stability (0.234), the lowest of all regimes — the archetypal "churning" trajectory.

Regime 4 — Employed Mid (n = 3,510, 12.9%): Near-universal employment (98.5%) with mixed income (low 26.0%, mid 31.8%, high 42.2%). Stability 0.407.

Regime 3 — Employment-Inactive Mix (n = 3,333, 12.2%): Mixed employment (58.8%) and inactivity (41.2%), zero unemployment. Income relatively balanced (low 20.0%, mid 39.6%, high 40.4%). Stability 0.280. Likely captures career breaks, part-time

work, and caring responsibilities.

Regime 6 — Low-Income Churn (n = 2,064, 7.6%): Employment (59.1%), unemployment (9.8%), and inactivity (31.1%), with the highest low-income rate (61.1%) of any regime. Stability 0.289. The second key disadvantaged regime: individuals cycle between low-paid employment, unemployment, and inactivity. Combined with Regime 2, these two regimes comprise 27.5% of the sample.

Regime 5 — High-Income Inactive (n = 1,813, 6.6%): Mixed employment (41.3%) and inactivity (58.3%), with high income (67.2%). Stability 0.400. Likely early retirees with substantial assets or high-income households with one inactive partner.

The correlation between regime stability and high-income rate is strong ($r = 0.77$, $p < 0.01$), confirming that the most stable trajectories tend to be the most advantaged.

4.4.1 H_0 Geometry vs. GMM Regimes

H_0 persistent homology encodes the Vietoris–Rips single-linkage merge tree. Cutting at $k = 7$ yields labels that are structurally distinct from GMM regimes:

ARI (H_0 tree $k = 7$ vs. GMM): 0.00004 **Component sizes:** C0: 99.98% (27,274 trajectories); C1–C6: singletons (1 trajectory each)

H_0 GMM	R0	R1	R2	R3	R4	R5	R6	Total
C0 (99.98%)	3,781	7,358	5,415	3,333	3,510	1,813	2,064	27,274
C1–C6 (singletons)	6	0	0	0	0	0	0	6

Trajectory space is topologically connected — a single giant H_0 component — with density substructure inside it (the 7 GMM regimes). The 6 outlier singletons are all from Mixed Churn (R0), which occupies the most diffuse region of embedding space. The near-zero ARI is expected: H_0 tracks global connectivity and boundary geometry, while GMM identifies interior density modes. H_0 and GMM operate on different geometric properties — global connectivity versus local density — and are complementary. The trajectory manifold is a single connected object with heterogeneous density.

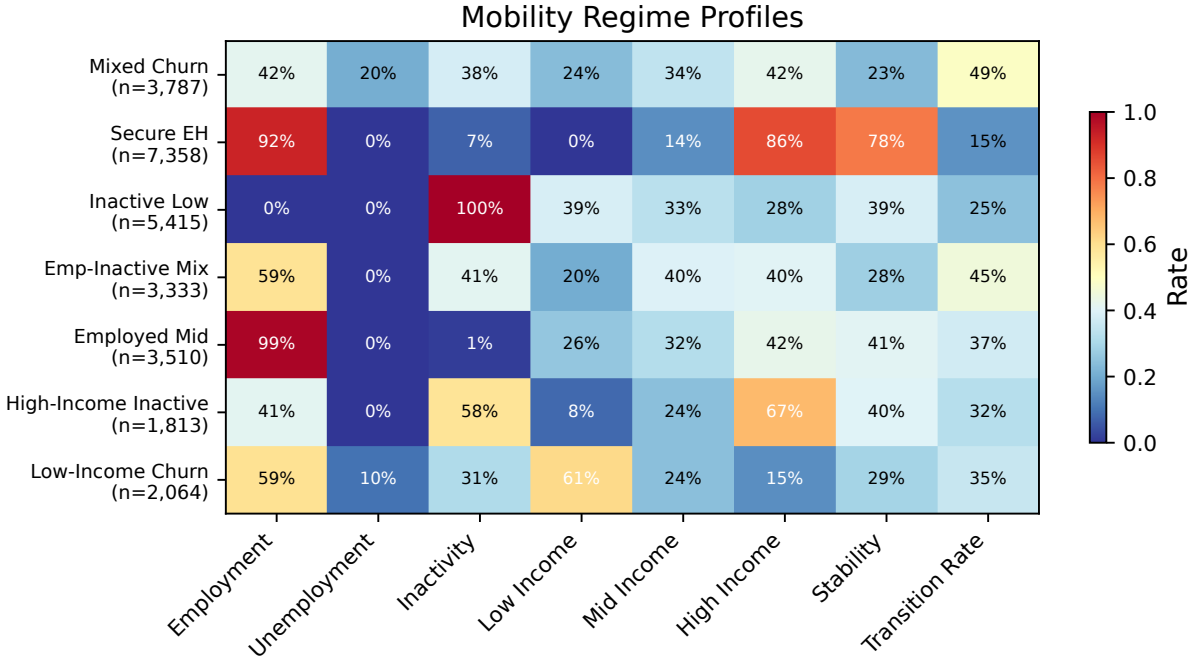

Figure 7: Regime profile heatmap. Rows are the seven mobility regimes; columns are labour-market metrics. Values are rates (0–1).

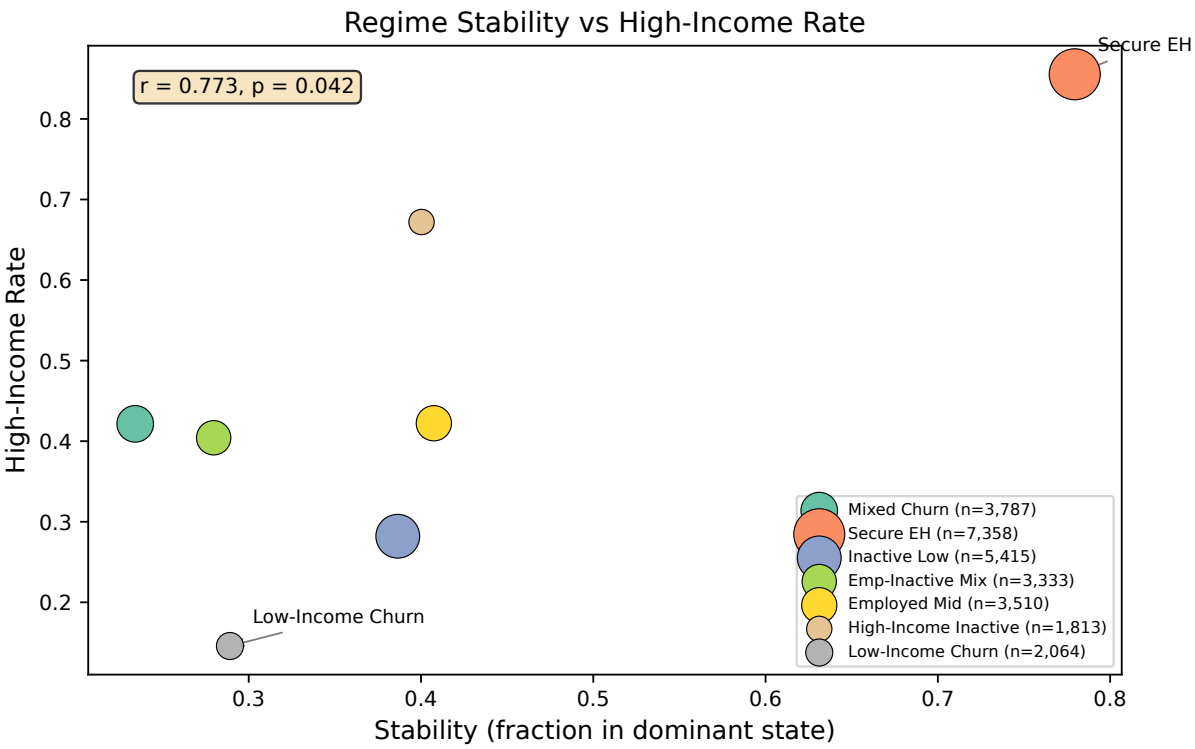

Figure 8: Stability versus high-income rate for each regime. Bubble size is proportional to regime membership.

4.4.2 Regime Persistence Over the Life Course

Overlapping 10-year windows (5-year step) for all 27,280 individuals yield 54,560 career-phase windows. The resulting 7 × 7 transition matrix is strongly diagonal-dominant:

	R0	R1	R2	R3	R4	R5	R6
R0	0.53	0.13	0.12	0.06	0.08	0.02	0.07
R1	0.03	**0.78**	0.03	0.03	0.08	0.04	0.02
R2	0.02	0.00	**0.97**	0.01	0.00	0.00	0.00
R3	0.05	0.14	0.18	0.41	0.15	0.04	0.04
R4	0.02	0.17	0.00	0.07	**0.66**	0.01	0.07
R5	0.05	0.14	0.48	0.06	0.01	0.26	0.00
R6	0.05	0.07	0.21	0.06	0.14	0.00	0.46

The strongest self-transitions occur in Regime 2 — Inactive Low (0.97), Regime 1 — Secure High-Employment (0.78), and Regime 4 — Employed Mid (0.66). The two disadvantaged regimes differ sharply in persistence: Inactive Low is near-absorbing (97%), while Low-Income Churn is moderately sticky (46%) with significant transitions to Inactive Low (21%) and Employed Mid (14%).

The 97% self-transition for Inactive Low likely reflects, in part, lifecycle inactivity: 86.1% of Inactive Low window-observations come from individuals aged ≥ 60.

Non-overlapping window validation. Non-overlapping windows (window_years=5, window_step=5) yield a self-transition rate of 0.601 versus 0.686 for overlapping windows (difference = 0.084). The escape rate from disadvantaged regimes is 11.0% under non-overlapping windows versus 5.6% under overlapping windows. The overlap-induced inflation is approximately 8.4 percentage points for self-transition. The core regime structure is preserved: the transition matrix remains diagonal-dominant and the qualitative ordering of regime persistence is unchanged.

Age-adjusted escape rates.

Group	n (disadvantaged start)	n (escaped)	Escape rate
Overall	7,453	416	5.6%
Working-age (< 60)	2,163	386	17.8%
Retirement-age ($\geq$ 60)	5,978	6	0.1%

The raw 5.6% overall escape rate is heavily confounded by the predominance of retirees in Inactive Low.

Logistic regression. The extended model (4,832 individuals with complete covariate data, pseudo-R^2 = 0.479):

Predictor	OR	95% CI	p-value
Age at first window	0.930	[0.883, 0.980]	0.006
Female	0.760	[0.558, 1.036]	0.083
Cohort 1950s (vs pre-1950)	7.029	[1.499, 32.96]	0.013
Cohort 1960s	23.825	[4.088, 138.8]	< 0.001
Cohort 1970s	16.273	[1.970, 134.4]	0.010
Cohort post-1980	8.204	[0.632, 106.5]	0.108
Initial regime 6 (vs regime 2)	20.556	[10.65, 39.69]	$< 10^{-19}$
NS-SEC Prof./Managerial	0.815	[0.477, 1.393]	0.456
NS-SEC Routine/Manual	0.839	[0.608, 1.157]	0.284

Each additional year of age reduces escape odds by 7% (OR = 0.93, p = 0.006). Birth cohort effects are dramatic: the 1960s cohort has 24 times the escape odds of the pre-1950 reference (OR = 23.8, p < 0.001). Initial regime is the strongest predictor: individuals starting in Regime 6 have 21 times the escape odds of those starting in Regime 2, the two disadvantaged regimes have very different mobility profiles. Parental NS-SEC is not significant after controlling for age, cohort, and initial regime.

Among the 7,453 individuals whose first window falls in a disadvantaged regime, only 416 (5.6%) ever reach an advantaged regime. All observed escapes occur in a single window-to-window transition — no multi-step "ladders" are observed, though the limited number of windows per person (mean 2.0) constrains our ability to detect longer escape paths.

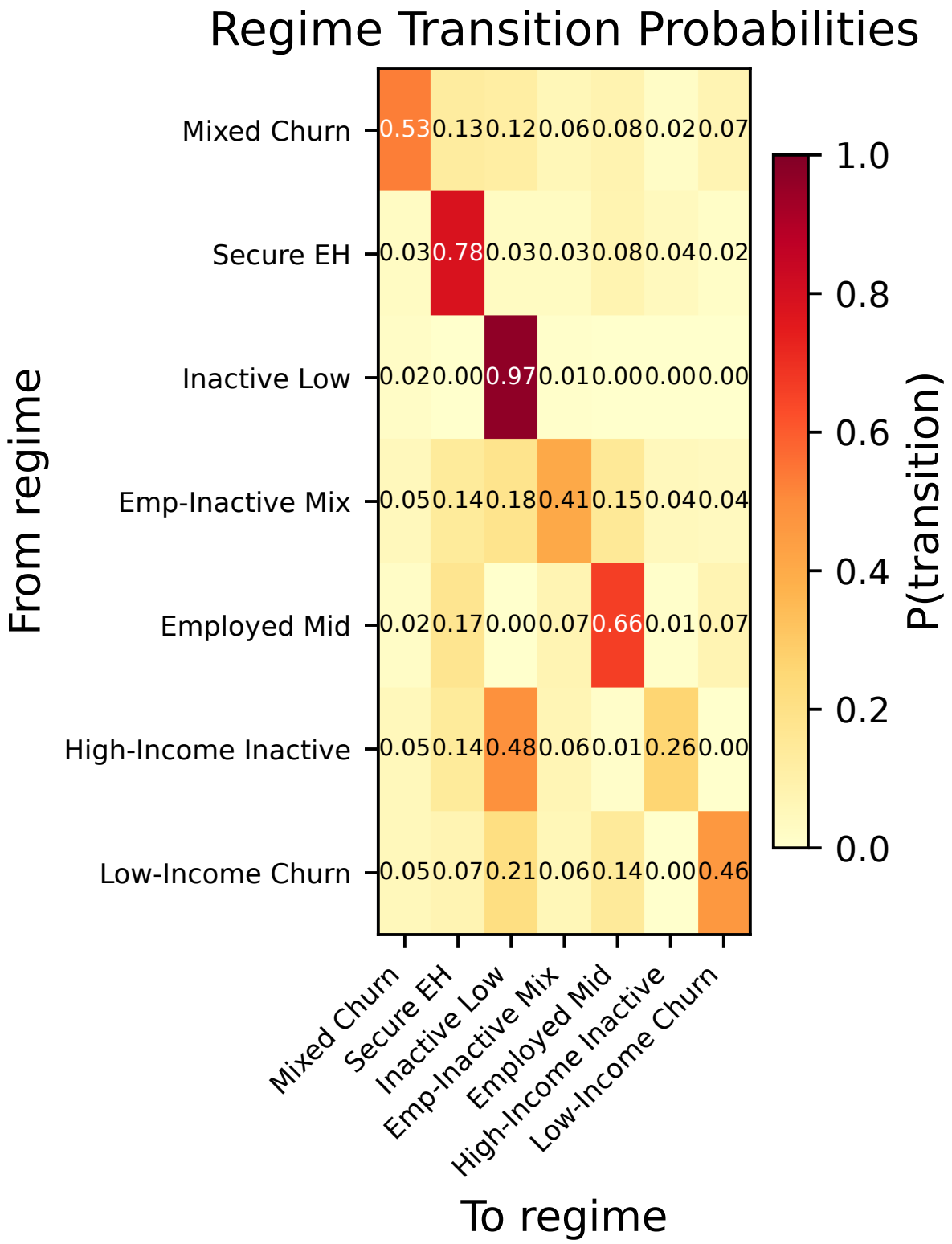

Figure 9: Regime transition probability matrix. Cell (i, j) shows $P(\text{regime}_{t+1} = j \mid \text{regime}_t = i)$ estimated from overlapping career-phase windows.

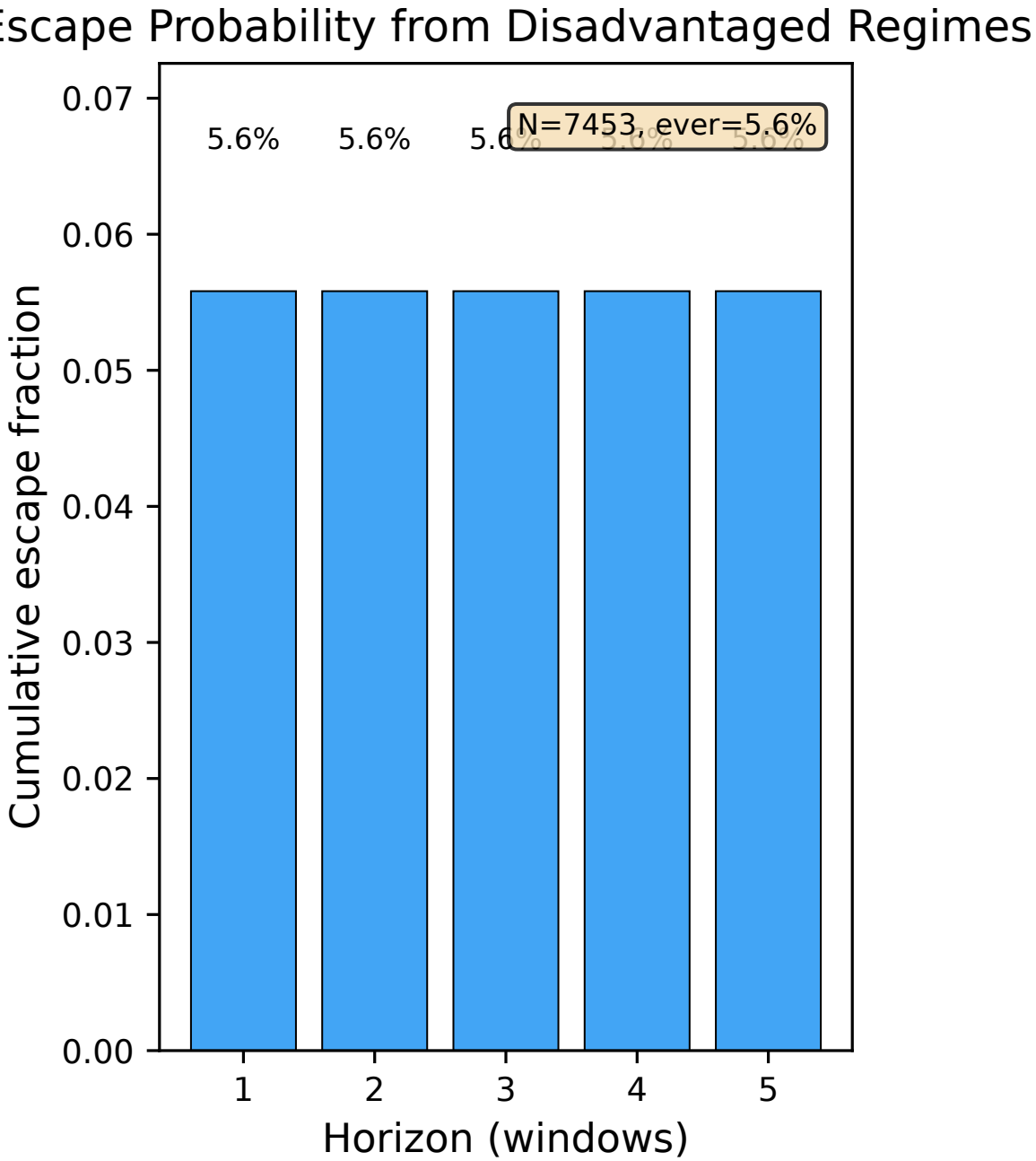

Figure 10: Cumulative escape probability from disadvantaged regimes (Regimes 2 & 6) by horizon window. Among $N = 7{,}453$ individuals starting in a disadvantaged regime, 5.6% ever reach an advantaged regime.

4.5 Cycle and Trap Analysis

The 5,962 H_1 features indicate substantial looping in the trajectory space, but the cycling is not topologically distinctive under either test statistic. Under total persistence, order-shuffle H_1 is significant at $L = 5{,}000$ (p = 0.020) but landmark-sensitive (non-significant at $L = 2{,}500$ and $L = 8{,}000$). Under Wasserstein distances, order-shuffle H_1 is clearly non-significant (p = 0.854), and Markov-1 H_1 is non-significant (p = 0.086). The H_1 null result is therefore robust across test statistics: the trajectory manifold lacks persistent cyclical structure beyond what first-order Markov dynamics produce.

H_1 loops correspond to cyclical oscillation patterns in the embedding space, not to poverty traps in the absorbing-state sense (§2.3). The most persistent loops (maximum persistence 3.21) involve cycling between low-income employment, unemployment, and inactivity — states concentrated in the Low-Income Churn and Mixed Churn regimes. These descriptive features may prove useful for identifying target populations even though they do not exceed Markov baselines.

4.5.1 Phase-Ordering Test

Using the PCA embedding of 50,000 overlapping windows with 3,000 maxmin landmarks and n = 500 permutations:

Dimension	Observed	Null mean ± std	p-value	95% CI
H_0	19,119.4	19,150.2 ± 17.2	0.98	[0.968, 0.992]
H_1	2,071.2	2,078.0 ± 19.9	0.542	[0.498, 0.586]

The main topological signal arises from the internal sequential structure of full trajectories (captured by bigrams in the embedding) rather than from the higher-order sequencing of phases across the life course.

4.6 Stratified Comparisons

Pairwise Wasserstein distances between group-specific persistence diagrams reveal significant topological differences by social origin, gender, and birth cohort. We con-

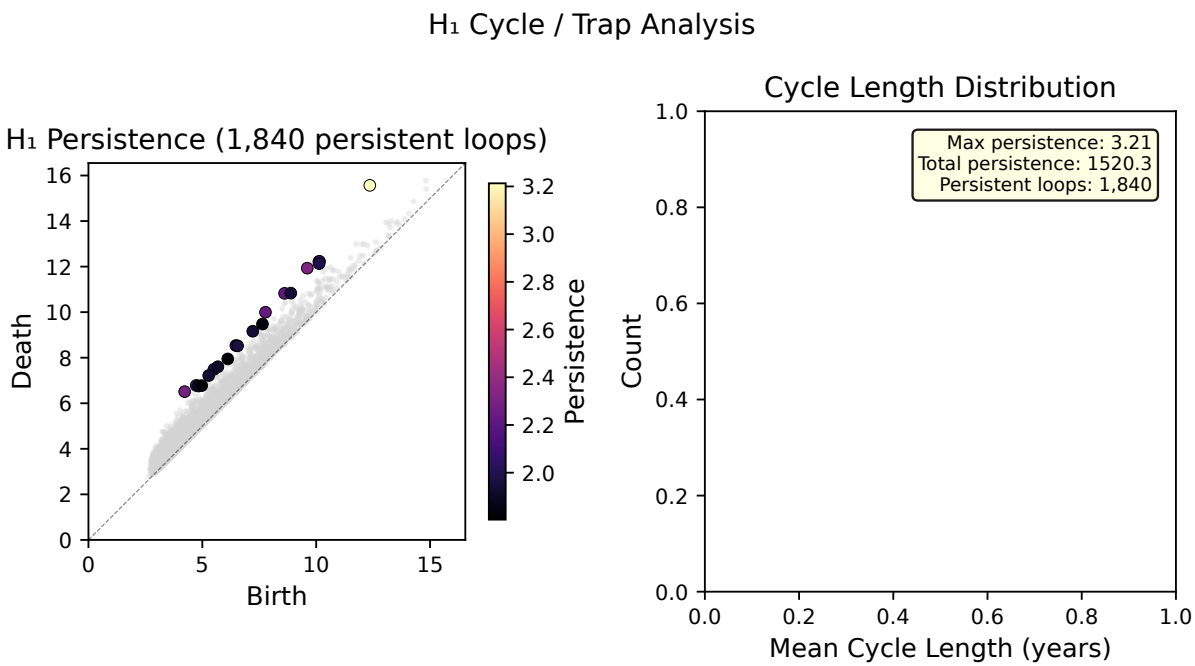

Figure 11: Cycle (H_1) analysis. Left: H_1 persistence diagram with top 20 features highlighted. Right: cycle-length distribution.

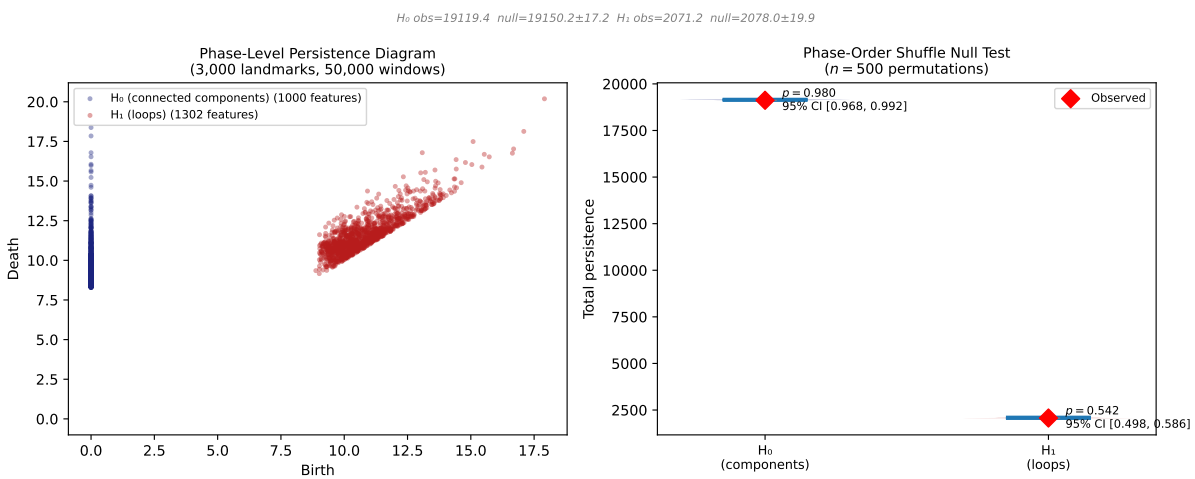

Figure 12: Phase-level persistent homology. Left: persistence diagram for career-phase embeddings ($L = 3{,}000$ landmarks, 50,000 windows). Right: phase-order shuffle null test ($n = 500$ permutations, 3,000 landmarks). The observed total persistence is not an outlier relative to the null (H_0: $p = 0.98$, 95% CI [0.968, 0.992]; H_1: $p = 0.542$, 95% CI [0.498, 0.586]).

duct 50 dimension-specific tests across three stratification axes and report both raw p-values and Benjamini-Hochberg (BH) FDR-corrected q-values.

Of the 50 tests, 30 are significant at $p < 0.05$ (raw). After BH FDR correction, all 30 remain significant at $q < 0.05$. Of these, 16/25 are H_0 tests and 14/25 are H_1 tests.

Gender. Male (n = 11,243) versus Female (n = 14,402): Wasserstein distance significant for H_0 ($p < 0.001$, BH $q < 0.001$; observed distance 33.7 vs. null mean 21.0), not for H_1 ($p = 0.27$). Men and women occupy different cluster structures — the gender gap in employment and inactivity produces distinct H_0 topology — but their cycling patterns are topologically similar.

Parental NS-SEC (*note: 39.1% of the sample has missing NS-SEC; this subsample is non-random, §3.1*). Three groups yield significant H_0 differences for all three pairwise comparisons ($p < 0.001$ each). The largest H_0 Wasserstein distance is between Professional/Managerial and Routine/Manual (114.8, vs. null mean 99.8). The sharper separation between extreme classes, with Intermediate occupying a genuinely intermediate topological position, aligns with the cumulative advantage hypothesis (DiPrete and Eirich, 2006).

Birth cohort. Of 42 cohort tests, 26 are significant at $p < 0.05$ (raw), all surviving BH correction. The 1930s cohort is topologically most distinctive.

Table 3: Stratified Wasserstein Test Summary (FDR-Corrected).

Axis	Tests	Significant (raw)	Significant (BH $q < 0.05$)
Gender	2	1	1
Parental NS-SEC	6	3	3
Birth cohort	42	26	26
Total	**50**	**30**	**30**

4.7 Cross-Era Validation: BHPS-Era Trajectories (1991–2008)

We construct 8,509 BHPS-era trajectories (mean length 14.5 years) using the fixed cross-wave identifier linkage (§3.1).

PH results. BHPS-era PH (L = 5,000, PCA-20D) yields H_0 total persistence =

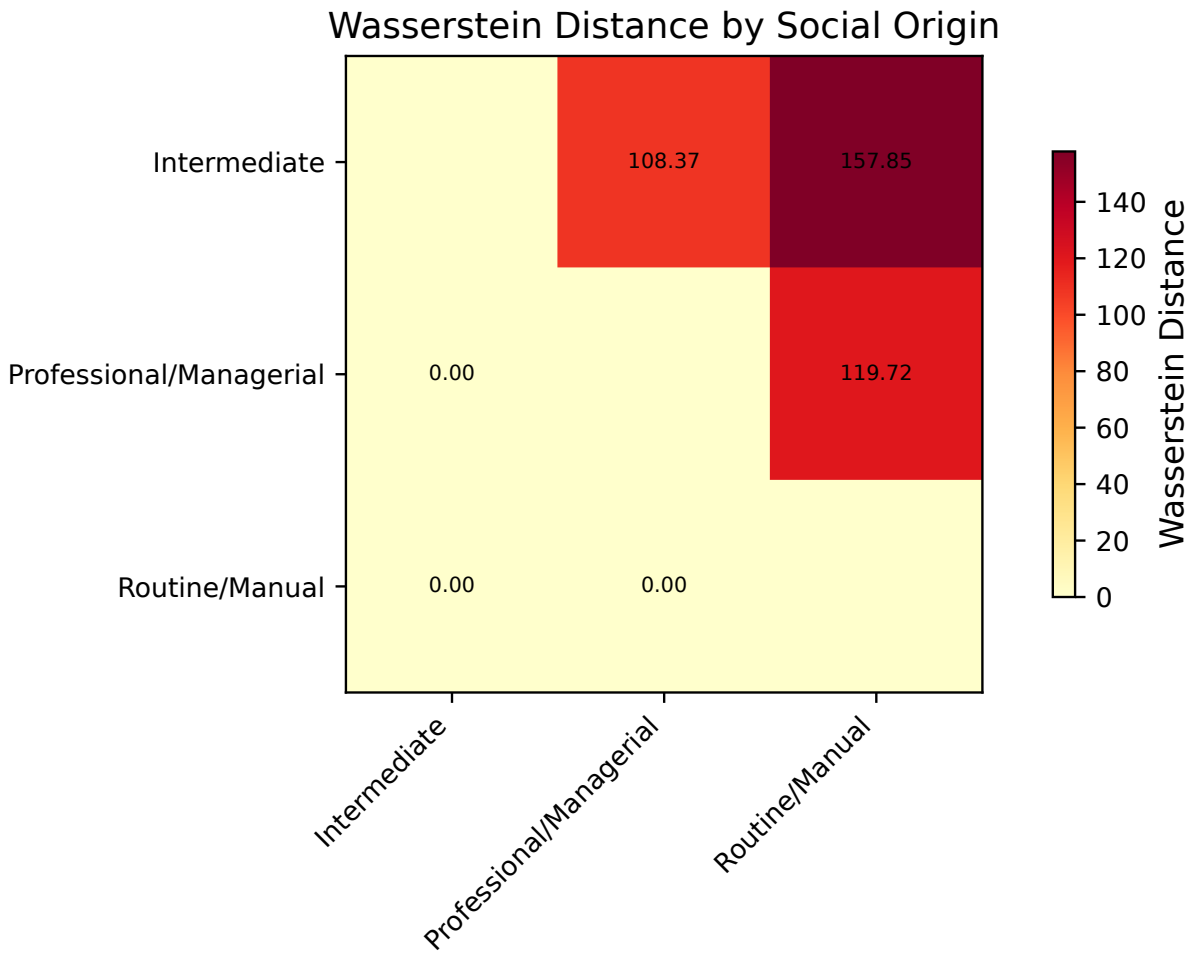

Figure 13: Pairwise Wasserstein distances between persistence diagrams by parental NS-SEC. Stars mark statistically significant differences.

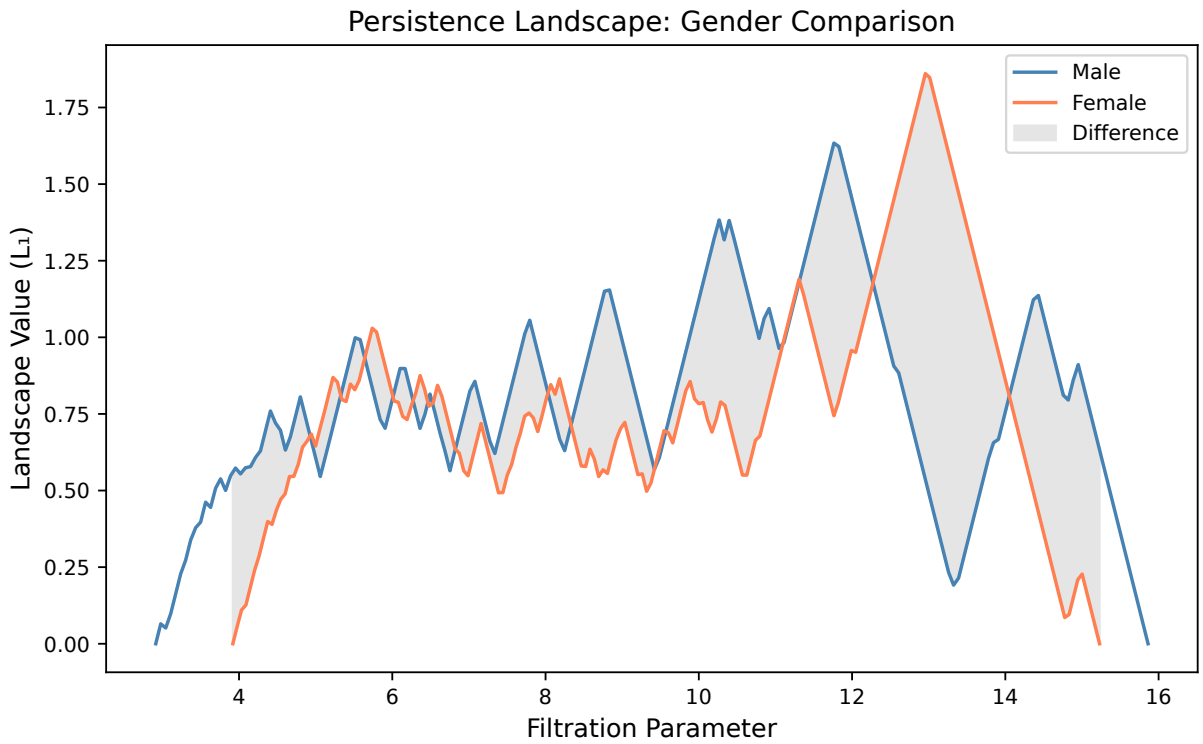

Figure 14: Persistence landscape comparison (L_1) for male versus female trajectories. Shaded region shows the difference.

10,763.2 and H_1 total persistence = 823.0 (1,114 persistent loops).

Null models (total persistence). The BHPS-era null battery ($n = 500$) reproduces the qualitative pattern:

Null Model	H_0 p	H_1 p
Order shuffle	**0.000**	0.520
Markov-1	1.000	1.000
Label shuffle	0.272	0.396

Order-shuffle H_0 is strongly rejected. Markov-1 is not rejected under total persistence, with the Markov null generating more persistence than observed (null mean H_0 = 12,753.8) — the same direction as USoc. The cross-era replication strengthens the total-persistence findings.

Wasserstein tests (BHPS). We run the Wasserstein battery on BHPS-era trajectories ($L = 2{,}000$, $n = 100$ permutations):

The BHPS order-shuffle Wasserstein result is strongly rejected ($p = 0.000$), with the mean obs-null distance (16.17) more than 2.5 times the null-null baseline (6.36). This

Table 4: BHPS-Era Wasserstein Results ($L = 2{,}000$, $n = 100$).

Null Model	Dim	W(obs↔null) $\mu \pm \sigma$	W(null↔null) $\mu \pm \sigma$	p	Sig?
Order shuffle	H_0	16.17 ± 1.60	6.36 ± 1.70	**0.000**	**yes**
Order shuffle	H_1	161.59 ± 3.47	155.66 ± 4.73	0.098	no
Markov-1	H_0	23.18 ± 2.22	7.05 ± 1.77	**0.000**	**yes**
Markov-1	H_1	149.11 ± 2.37	134.24 ± 3.69	**0.000**	**yes**

is a clear strengthening relative to the USoc result (p = 0.058) — the smaller BHPS-era sample (8,509 vs 27,280 trajectories) produces a cleaner separation, possibly because the longer mean trajectory length (14.5 vs 12.9 years) provides richer temporal structure for the n-gram embedding to exploit. H_1 remains non-significant (p = 0.098), consistent with the USoc pattern.

The BHPS Markov-1 Wasserstein result provides the strongest cross-era validation: both H_0 (p = 0.000) and H_1 (p = 0.000) are decisively rejected, with the mean H_0 obs-null distance (23.18) more than three times the null-null baseline (7.05). The BHPS Markov-1 H_1 rejection (p = 0.000) is a new finding: the USoc Markov-1 H_1 was borderline (p = 0.086), but the BHPS-era data decisively rejects it. The longer BHPS trajectory lengths provide greater statistical power — but the substantive implication is that BHPS-era loop structure also exceeds Markov expectations under diagram-level comparison, even though total persistence says otherwise. The test-statistic discrepancy replicates cross-era in both homological dimensions.

Regime structure. BIC selects $k = 8$ on BHPS-era embeddings, one regime more than USoc's $k = 7$. Core types are comparable: stable employed high-income (stability = 0.83), economically inactive low-income (employment = 0.0%, stability = 0.44), and multiple churning/mixed regimes.

Spanning trajectories. 8,459 spanning trajectories (observed in both surveys) yield order-shuffle H_0 p = 0.000 and Markov-1 H_0 p = 1.000 (total persistence), confirming the same pattern.

5 Discussion

5.1 What TDA Adds Beyond Sequence Analysis

PH provides a geometric characterisation of the trajectory point cloud unavailable from pairwise dissimilarity methods. The persistence diagrams encode the multi-scale connectivity (H_0) and cyclical structure (H_1) of the entire trajectory space: the most persistent H_0 component (15.81) is nearly four times the mean (4.08), indicating a few well-separated regimes with many weakly separated sub-clusters. The near-zero ARI between H_0 components and GMM assignments (§4.4.1) demonstrates that H_0 PH and GMM operate on different geometric properties — global connectivity versus local density — and are complementary analyses.

The Markov memory ladder provides a framework for bounding the memory-complexity of observed topology. That the Markov conclusion depends on the test statistic (§4.3) is the methodologically important finding: scalar persistence summaries can mask structural differences that Wasserstein comparisons detect.

OM-based clustering at matched $k = 7$ yields ARI = 0.26: the Secure High-Employment regime is identified nearly identically (profile correlation = 0.97), while Mixed Churn diverges most (correlation = 0.02). The broad regime typology is recoverable by either method; TDA adds the geometric characterisation and formal testing.

5.2 The Markov Memory Ladder and the Test-Statistic Discrepancy

The Markov memory ladder is our primary methodological contribution. The central finding is that the Markov conclusion depends on how persistence diagrams are compared. Under total persistence, Markov-1 surrogates generate *more* aggregate persistence than observed ($p = 1.000$), suggesting the topology is "bounded" by first-order dynamics. Under Wasserstein distances, the observed H_0 diagram is significantly different from Markov-1 null diagrams ($p = 0.002$), indicating that the specific configuration of connected components — which clusters form, at what scale, and in what hierarchical relationship — carries structural information that a first-order Markov

chain cannot reproduce.

We interpret this discrepancy as follows. The globally estimated Markov transition matrix generates trajectories that, in aggregate, are more diffuse than the observed data — they spread more widely in embedding space and produce more topological features. But the pattern of those features differs from what the real trajectories produce. The real data has regime-specific substructure: clusters that form at particular filtration scales and merge in particular ways. The Markov process, which uses a single global transition matrix without regime-specific parameters, smears out this substructure. The total persistence is higher (more features, longer total lifetime) but the configuration of the diagram — the distribution of birth-death pairs — is different.

This interpretation is consistent with the intra-regime compactness results (§4.3): all seven regimes are significantly more compact in the observed embedding than under order-shuffled re-embeddings, with the churning regimes showing the largest compactness ratios. The trajectory data has structural regularity within regimes that a global Markov process does not capture.

These results come with caveats that bear on interpretation. The Wasserstein test uses $L = 2{,}000$ landmarks rather than the $L = 5{,}000$ used for total-persistence tests, because computing persistence diagrams for each null realisation is expensive; the landmark count difference could affect the comparison, though total-persistence results are stable across L = 2,500 to 8,000 (Appendix B). The Wasserstein battery covers label shuffle, order shuffle, Markov-1, and Markov-2 nulls but omits cohort shuffle due to metadata constraints — its total-persistence result (p = 0.405) already establishes it as a negative control alongside label shuffle. More fundamentally, the Markov-1 Wasserstein rejection does not specify *what kind* of non-Markovian structure is present — it could reflect individual-level heterogeneity, duration dependence, or higher-order memory. Richer null models (hidden Markov, latent-type, variable-order) would help decompose this.

The Markov memory ladder is transportable: it can be applied to any TDA analysis of sequential data. The lesson from our analysis is that the ladder should be climbed

with both scalar and diagram-level test statistics, since they can give different answers.

5.3 H_1 Cycles: A Qualified Negative Result

H_1 loops correspond to cyclical oscillation patterns in the embedding space rather than to poverty traps in the absorbing-state sense. In the USoc data, H_1 is non-significant under Wasserstein distances for both order-shuffle (p = 0.854) and Markov-1 (p = 0.086). The landmark-sensitive order-shuffle H_1 significance under total persistence (p = 0.020 at L = 5,000, non-significant at other L) does not replicate under Wasserstein testing, reinforcing a cautious interpretation.

However, the BHPS-era data produces a stronger signal: Markov-1 H_1 is decisively rejected under Wasserstein (p = 0.000), with the mean obs-null distance (149.11) far above the null-null baseline (134.24). The BHPS trajectories — longer on average (14.5 vs 12.9 years) and drawn from a different labour-market era — show cyclical structure that exceeds Markov expectations at the diagram level. Whether this reflects genuine non-Markovian cycling patterns in the 1990s–2000s economy or is an artefact of the longer window length is a question that can be directly addressed: truncating BHPS trajectories to 12–13 years before running the Wasserstein H_1 test would discriminate between an era effect and a window-length artefact. The USoc H_1 borderline (p = 0.086) is suggestive of the same effect, weakened by shorter trajectories.

This does not mean that people do not cycle between employment and poverty in the USoc era. They manifestly do — the Low-Income Churn and Mixed Churn regimes demonstrate extensive state cycling. Rather, the USoc null result means that the topological shape of this cycling is predictable from one-step transition probabilities in that sample. The BHPS result suggests that with longer observation windows, the diagram-level structure of cycling may diverge from Markov baselines.

This qualified negative result directs analytical attention toward the stronger H_0 results and the escape-rate analysis (§4.4.2), while flagging H_1 as worthy of further investigation with longer trajectories.

5.4 Limitations

Spatial dimension. The analysis is national and non-spatial. The Markov memory ladder is directly applicable to regional comparisons, which we identify as a priority extension.

State space discretisation. The nine-state space, while richer than most sequence analysis applications (typically 3–5 states), is a simplification. The 60%/120% thresholds are conventional but arbitrary; sensitivity to alternatives is not examined.

Embedding sensitivity. PCA-20D is the primary embedding. The UMAP-16D check yields ARI = 0.33 against PCA at $k = 7$. We test a grid of six configurations varying n-gram order, weighting, and PCA dimensionality:

Configuration	H_0 p	H_1 p	H_0 z	H_1 z	n
Bigram TF, PCA-20 (baseline)	< 0.001	0.040	+15.8	+1.6	50
Bigram TF-IDF, PCA-20	< 0.001	1.000	+6.5	-2.6	20
Trigram TF, PCA-20	< 0.001	1.000	+8.1	-7.6	20
Trigram TF-IDF, PCA-20	< 0.001	< 0.001	+23.6	+4.0	20
Bigram TF, PCA-10	< 0.001	0.920	+2.6	-1.6	50
Bigram TF, PCA-30	< 0.001	0.840	+16.5	-1.3	50

H_0 is uniformly significant across all six configurations, confirming that the connected-component structure is robust. H_1 is embedding-sensitive: significant for trigram TF-IDF but not for most others. We report TF (not TF-IDF) as the primary weighting because TF preserves raw transition frequencies without downweighting common state transitions; frequently recurring transitions such as EH→EH are substantively meaningful (they capture regime stability) and should not be treated as noise. H_1 embedding-sensitivity is therefore expected and does not undermine the primary H_0 results.

H_0 **chaining.** Single-linkage chaining limits balanced H_0-based clustering in high-dimensional VR complexes: the merge tree produces one dominant component (99.98%) plus outlier singletons, yielding ARI ≈ 0 against GMM. This is a known property of VR H_0 in high dimensions (Bobrowski and Kahle, 2018) and motivates future explo-

ration of density-aware TDA methods (e.g., DTM filtrations, Mapper).

Landmark sampling. Maxmin landmark selection (5,000 from 27,280) preserves global geometry but may undersample rare trajectory types. Results are stable across L = 2,500 to 8,000. The Wasserstein tests use $L = 2{,}000$ due to computational cost; validating at higher L is future work.

Age and lifecycle confounding. The near-absorbing Inactive Low regime (97% self-transition) partly reflects retirement. We address this via age-stratified analysis (§4.4.2): 86.1% of Inactive Low observations come from individuals aged ≥ 60. Working-age escape rates (17.8%) are more than three times the raw rate (5.6%).

Markov null limitations. Our Markov nulls test against globally estimated Markov dynamics. They do not account for individual-level heterogeneity or duration dependence. The Wasserstein rejection of Markov-1 (p = 0.002) suggests that such heterogeneity may be precisely what the test is detecting — the global Markov chain cannot reproduce the regime-specific substructure of the observed diagrams. Stratified Markov nulls (separate transition matrices per regime) could help disambiguate this. The per-regime stratified test under total persistence (§5.2 in the previous draft) found non-rejection, but the effective stratification was limited (one regime contained 99.9% of trajectories at the point-cloud level). Stratified Wasserstein testing is future work.

Family dimension. We analyse employment-income trajectories only; partnership and household composition are not included.

Imputation. Forward-fill imputation for gaps of up to two years affects 4.9% of person-years.

Cross-survey harmonisation. BHPS and USoc use different sampling frames, questionnaire designs, and income measurement methodologies. Harmonisation artefacts may contribute to cross-era topological differences. The BHPS-era Wasserstein results (§4.7) confirm the cross-era replication of the test-statistic discrepancy.

Pipeline sensitivity. We vary minimum trajectory length (8, 10, 12 years), weighting (TF vs TF-IDF), and filtration threshold (50th, 75th, 90th percentile). Under TF, results are highly stable: k = 7 at 10/12 years, k = 8 at 8 years; H_0 persistence varies <

6%; threshold has no effect on H_0 and negligible effect on H_1. TF-IDF reshapes the embedding substantially (ARI = 0.466 against TF baseline, 93% drop in total persistence), confirming that the choice of TF weighting is consequential.

5.5 Robustness Summary

Table 5: Robustness checks summary.

Check	Status	Summary
Wasserstein diagram tests	Expanded (§4.3)	Markov-1 H_0 **rejected** (p = 0.002); order-shuffle H_0 borderline (p = 0.058); label shuffle negative control (p = 0.452); H_1 non-significant; Markov-2 H_0 not rejected (p = 0.546), H_1 not rejected (p = 0.078)
Total-persistence tests	Completed (§4.3)	Order-shuffle H_0 $p < 0.005$; Markov-1 p = 1.000; H_1 landmark-sensitive
Test-statistic discrepancy	New (§4.3, §5.2)	Total persistence and Wasserstein give opposite Markov-1 conclusions; scalar masks diagram-shape differences
UMAP-16D embedding	Completed (§3.2)	ARI = 0.33; order-shuffle H_0 $p < 0.001$; mirrors PCA pattern
Landmark count (L = 2,500/5,000/8,000)	Completed (Appendix B)	Order-shuffle H_0 $p < 0.001$ at all L; Markov-1 non-significant at all L (total persistence)
Embedding sensitivity grid	Completed (§5.4)	6 configs: H_0 $p < 0.001$ uniformly; H_1 embedding-sensitive
Non-overlapping windows	Completed (§4.4.2)	Escape rate 5.6% vs 11.0%; core structure preserved
OM/TraMineR baseline	Completed (§3.7, §5.1)	ARI = 0.26 at k = 7
Wasserstein FDR correction	Completed (§4.6)	30/50 significant raw; all 30 survive BH
Phase-ordering null (n = 500)	Completed (§4.5)	H_0 p = 0.98; H_1 p = 0.542
GMM bootstrap stability	Completed (§3.5)	ARI = 0.646 $\pm$ 0.086
Bootstrap null stability	Completed (§4.3)	10/10 order-shuffle significant; 10/10 Markov-1 significant; no flips
Intra-regime compactness	Completed (§4.3)	All 7 regimes $z = -30$ to -68, all $p < 0.001$
NS-SEC missingness	Completed (§3.1)	39.1% missing; non-random
BHPS cross-era replication	Expanded (§4.7)	Same qualitative pattern (total persistence); Wasserstein order-shuffle H_0 p = 0.000; Markov-1 H_0 p = 0.000, H_1 p = 0.000
Age-adjusted escape rates	Completed (§4.4.2)	Working-age escape 17.8%; pseudo-R^2 = 0.479
Pipeline sensitivity	Completed (§5.4)	TF stable; TF-IDF reshapes; threshold no effect

6 Conclusion

The test statistic you choose determines whether UK career trajectories look Markovian.

Under total persistence — the scalar summary standard in TDA hypothesis testing — Markov-1 surrogates generate more aggregate persistence than the observed data (p = 1.000), and the natural conclusion would be that first-order transition dynamics fully account for the topology of the trajectory manifold. Under Wasserstein diagram distances, the same null is rejected (p = 0.002): the observed H_0 diagram is an outlier relative to the Markov null cloud. The Markov chain overshoots the total but distributes persistence differently across birth-death pairs — it smears out regime-specific substructure that the real trajectories preserve. This discrepancy emerged from the dual-statistic design — comparing both statistics on the same null battery — and replicates cross-era on 8,509 BHPS trajectories (1991–2008) for both H_0 and H_1. Had we used only the standard scalar test, as much of the TDA hypothesis-testing literature does, we would have declared the topology Markov-bounded and moved on.

The substantive upshot is that the seven mobility regimes identified here — from Secure High-Employment (27%, stability 0.78) to Low-Income Churn (7.6%, stability 0.29) — are not merely density modes in the embedding space. Their hierarchical connectivity structure, encoded in the pattern of H_0 birth-death pairs, carries information about the data-generating process that cannot be recovered from a globally estimated transition matrix. The regime landscape has internal geometry. That geometry sits at the second-order Markov boundary: Markov-2 reproduces the observed diagram configuration (p = 0.546), localising the non-Markovian signal to two-step memory rather than higher-order or indefinitely long dependence. For policy, this means that the regime structure is shaped by something richer than one-step transition probabilities — recent trajectory history matters — but it does not require modelling arbitrarily deep path dependence.

The H_1 result introduces a complication that future work will need to resolve. USoc-era loops are non-significant under all nulls and both statistics, suggesting that

cyclical trapping is predictable from Markov dynamics in shorter observation windows. But the longer BHPS trajectories (mean 14.5 vs 12.9 years) decisively reject Markov-1 H_1 under Wasserstein ($p = 0.000$). The trajectory manifold's cyclical structure may require longer windows to separate from Markov baselines, or the 1991–2008 labour market may have generated genuinely different cycling dynamics. Either way, the USoc H_1 negative result is weaker than it appears: it is bounded by trajectory length, not by the absence of cycling.

The 5.6% escape rate from disadvantaged regimes — 17.8% for working-age individuals after removing the retirement confound — quantifies what decades of panel-data research on poverty dynamics has described qualitatively: disadvantaged trajectories are sticky, the stickiness is regime-specific (Inactive Low is near-absorbing at 97%; Low-Income Churn permits 20× higher escape odds), and the strongest predictor of escape is which disadvantaged regime you start in, not your class origin (NS-SEC is non-significant after conditioning on age, cohort, and initial regime). The topological lens adds the observation that phase ordering across the life course contributes no additional topological structure ($p = 0.98$): the complexity of career dynamics lives inside each trajectory's sequential structure, not in the higher-order sequencing of career phases. The action, such as it is, happens locally.

Where does this leave TDA as a tool for life-course research? The regime typology itself is recoverable by optimal matching (ARI = 0.26 at matched $k = 7$), and the broad distinction between secure, churning, and inactive trajectories is visible from either method. TDA's contribution is the null-testing infrastructure and the geometric characterisation — and, as this paper demonstrates, the sensitivity of conclusions to how that geometry is measured. The Markov memory ladder is portable: it can be applied to any TDA analysis of sequential data, in any domain, with both scalar and diagram-level statistics. The lesson is to climb the ladder with both.

Immediate extensions include Mapper analysis of within-regime heterogeneity (Paper 2), zigzag persistence across annual cohort snapshots (Paper 3), and regional stratification at NUTS2 or Travel-to-Work Area level. Cross-national replication on SOEP

and PSID will test whether the regime structure reflects UK-specific labour market institutions or a more general pattern in the topology of career dynamics across welfare states.

A H_0 Merge Tree Analysis

The H_0 persistent homology on 5,000 maxmin landmarks encodes the single-linkage merge tree of the trajectory point cloud.

Top 10 H_0 lifetimes (persistence = death - birth, all born at 0): 15.81, 15.50, 14.90, 14.40, 14.16, 14.01, 13.62 (7th), 13.22 (8th), 13.15, 12.92

The ratio between the 7th and 8th most persistent features is 1.03×, indicating no sharp topological gap at $k = 7$.

Table S2: H_0 Component–GMM Regime Contingency (at $k = 7$ cut)

	R0 (Churn)	R1 (EH)	R2 (IL)	R3 (EI Mix)	R4 (EM)	R5 (IH)	R6 (LI Churn)	Total
C0 (99.98%)	3,781	7,358	5,415	3,333	3,510	1,813	2,064	27,274
C1–C6 (singletons)	6	0	0	0	0	0	0	6

ARI (H_0 $k = 7$ vs. GMM) = 0.00004.

Trajectory space is topologically connected: a single giant component with heterogeneous density. The 6 outlier singletons are all from Mixed Churn (R0). H_0 tracks global connectivity; GMM identifies local density modes. These measure different aspects of point-cloud geometry.

B Landmark Sensitivity

We recompute persistent homology at L = 2,500, 5,000, and 8,000 landmarks (approximately 9%, 18%, and 29% of the 27,280-point cloud). For each, we re-run order-shuffle and Markov-1 null tests (n = 50–100 per condition).

OS = order shuffle; M1 = Markov-1 (total persistence). n = number of finite features.

Total persistence scales approximately linearly with L, as expected. **Maximum persistence** is highly stable (H_0 max 15.50–15.81; H_1 max 3.21 at all three values).

Table 6: Landmark Sensitivity Analysis.

L	H_0 total	H_0 max	H_0 n	H_1 total	H_1 max	H_1 n	OS H_0 p	OS H_1 p	M1 H_0 p	M1 H_1 p
2,500	14,227.8	15.809	2,499	1,520.3	3.213	3,249	< 0.001	0.20	0.15	0.69
5,000	20,411.1	15.809	4,999	2,224.7	3.213	5,961	< 0.001	0.02	1.00	1.00
8,000	24,313.8	15.496	7,998	2,619.3	3.213	8,357	< 0.001	0.14	1.00	1.00

Null model p-values are qualitatively stable:

- Order-shuffle H_0: $p < 0.001$ at all L.
- Markov-1 H_0: Non-significant at all L under total persistence.
- Order-shuffle H_1: Landmark-sensitive; significant only at $L = 5{,}000$.

These results confirm that the key total-persistence findings are stable across a threefold range of landmark density. Wasserstein landmark sensitivity is a priority for future validation.

References

Abbott, A. (1995). Sequence analysis: New methods for old ideas. *Annual Review of Sociology*, 21:93–113.

Abbott, A. and Tsay, A. (2000). Sequence analysis and optimal matching methods in sociology. *Sociological Methods & Research*, 29(1):3–33.

Atienza, N., Gonzalez-Diaz, R., and Rucco, M. (2019). Persistent entropy for separating topological features from noise in Vietoris–Rips complexes. *Journal of Intelligent Information Systems*, 52(3):637–655.

Azariadis, C. and Stachurski, J. (2005). Poverty traps. In Aghion, P. and Durlauf, S., editors, *Handbook of Economic Growth*, volume 1, pages 295–384. Elsevier.

Barrett, C. B. and Carter, M. R. (2013). The economics of poverty traps and persistent poverty: Empirical and policy implications. *Journal of Development Studies*, 49(7):976–990.

Bauer, U. (2021). Ripser: Efficient computation of Vietoris–Rips persistence barcodes. *Journal of Applied and Computational Topology*, 5:391–423.

Blanden, J., Goodman, A., Gregg, P., and Machin, S. (2004). Changes in intergenerational mobility in Britain. In Corak, M., editor, *Generational Income Mobility in North America and Europe*, pages 122–146. Cambridge University Press.

Blanden, J., Gregg, P., and Macmillan, L. (2013). Intergenerational persistence in income and social class: The effect of within-group inequality. *Journal of the Royal Statistical Society: Series A*, 176(2):541–563.

Bobrowski, O. and Kahle, M. (2018). Topology of random geometric complexes: A survey. *Journal of Applied and Computational Topology*, 1:331–364.

Bukodi, E. and Goldthorpe, J. H. (2019). *Social Mobility and Education in Britain*. Cambridge University Press.

Carlsson, G. (2009). Topology and data. *Bulletin of the American Mathematical Society*, 46(2):255–308.

Cohen-Steiner, D., Edelsbrunner, H., and Harer, J. (2007). Stability of persistence diagrams. *Discrete & Computational Geometry*, 37(1):103–120.

Coulter, R., van Ham, M., and Findlay, A. M. (2016). Re-thinking residential mobility: Linking lives through time and space. *Progress in Human Geography*, 40(3):352–374.

de Silva, V. and Carlsson, G. (2004). Topological estimation using witness complexes. In Alexa, M. and Rusinkiewicz, S., editors, *Eurographics Symposium on Point-Based Graphics*, pages 157–166.

DiPrete, T. A. and Eirich, G. M. (2006). Cumulative advantage as a mechanism for inequality: A review of theoretical and empirical developments. *Annual Review of Sociology*, 32:271–297.

Edelsbrunner, H. and Harer, J. (2010). *Computational Topology: An Introduction*. American Mathematical Society.

Edelsbrunner, H., Letscher, D., and Zomorodian, A. (2002). Topological persistence and simplification. *Discrete & Computational Geometry*, 28(4):511–533.

Elzinga, C. H. and Liefbroer, A. C. (2007). De-standardization of family-life trajectories of young adults: A cross-national comparison using sequence analysis. *European Journal of Population*, 23(3–4):225–250.

Gauthier, J.-A., Widmer, E. D., Bucher, P., and Notredame, C. (2010). Multichannel sequence analysis applied to social science data. *Sociological Methodology*, 40(1):1–38.

Gidea, M. and Katz, Y. (2018). Topological data analysis of financial time series: Landscapes of crashes. *Physica A: Statistical Mechanics and its Applications*, 491:820–834.

Goldthorpe, J. H. (2016). Social class mobility in modern Britain: Changing structure, constant process. *Journal of the British Academy*, 4:89–111.

Goldthorpe, J. H. and Mills, C. (2008). Trends in intergenerational class mobility in modern Britain: Evidence from national surveys, 1972–2005. *National Institute Economic Review*, 205:83–100.

Halpin, B. and Chan, T. W. (1998). Class careers as sequences: An optimal matching analysis of work-life histories. *European Sociological Review*, 14(2):111–130.

Hiraoka, Y., Nakamura, T., Hirata, A., et al. (2016). Hierarchical structures of amorphous solids characterized by persistent homology. *Proceedings of the National Academy of Sciences*, 113(26):7035–7040.

HM Government (2022). Levelling up the United Kingdom. White Paper CP 604, HM Government.

Iacopini, I., Petri, G., Barrat, A., and Latora, V. (2019). Simplicial models of social contagion. *Nature Communications*, 10:2485.

Jenkins, S. P. (2011). *Changing Fortunes: Income Mobility and Poverty Dynamics in Britain*. Oxford University Press.

Lesnard, L. (2010). Setting cost in optimal matching to uncover contemporaneous socio-temporal patterns. *Sociological Methods & Research*, 38(3):389–419.

McInnes, L., Healy, J., and Melville, J. (2018). UMAP: Uniform manifold approximation and projection for dimension reduction. *arXiv preprint*.

Rizvi, A. H., Camara, P. G., Kandror, E. K., et al. (2017). Single-cell topological RNA-seq analysis reveals insights into cellular differentiation and development. *Nature Biotechnology*, 35(6):551–560.

Robette, N. and Thibault, N. (2008). Comparing qualitative harmonic analysis and optimal matching: An exploratory study of occupational trajectories. *Population-E*, 63(4):621–646.

Robinson, A. and Turner, K. (2017). Hypothesis testing for topological data analysis. *Journal of Applied and Computational Topology*, 1:241–261.

Saggar, M., Sporns, O., Gonzalez-Castillo, J., et al. (2018). Towards a new approach to reveal dynamical organization of the brain using topological data analysis. *Nature Communications*, 9:1399.

Savage, M., Devine, F., Cunningham, N., et al. (2013). A new model of social class? Findings from the BBC's Great British Class Survey experiment. *Sociology*, 47(2):219–250.

Singer, B. and Spilerman, S. (1976). The representation of social processes by Markov models. *American Journal of Sociology*, 82(1):1–54.

Sizemore, A. E., Phillips-Cremins, J. E., Ghrist, R., and Bassett, D. S. (2018). The importance of the whole: Topological data analysis for the network neuroscientist. *Network Neuroscience*, 3(3):656–673.

Stolz, B. J., Harrington, H. A., and Porter, M. A. (2017). Persistent homology of time-dependent functional networks constructed from coupled time series. *Chaos*, 27(4):047410.

Studer, M. and Ritschard, G. (2016). What matters in differences between life trajectories: A comparative review of sequence dissimilarity measures. *Journal of the Royal Statistical Society: Series A*, 179(2):481–511.

Tong, H. (1990). *Non-linear Time Series: A Dynamical System Approach*. Oxford University Press.

Turner, K., Mileyko, Y., Mukherjee, S., and Harer, J. (2014). Fréchet means for distributions of persistence diagrams. *Discrete & Computational Geometry*, 52(1):44–70.

Vandecasteele, L. (2010). Poverty trajectories, triggering events and exit routes. *Sociological Research Online*, 15(4):1–17.

Zomorodian, A. and Carlsson, G. (2005). Computing persistent homology. *Discrete & Computational Geometry*, 33(2):249–274.